The Single-Boom Crane Operation and Maintenance Manual

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Contents

1 Introduction

1.1 The Single-Boom Crane

This manual presents information about the single-boom crane system in the mechatronics laboratory at the University of Agder. The system consists of the mechanical structure, the original load-sensing hydraulic system, and the self-contained hydraulic system.

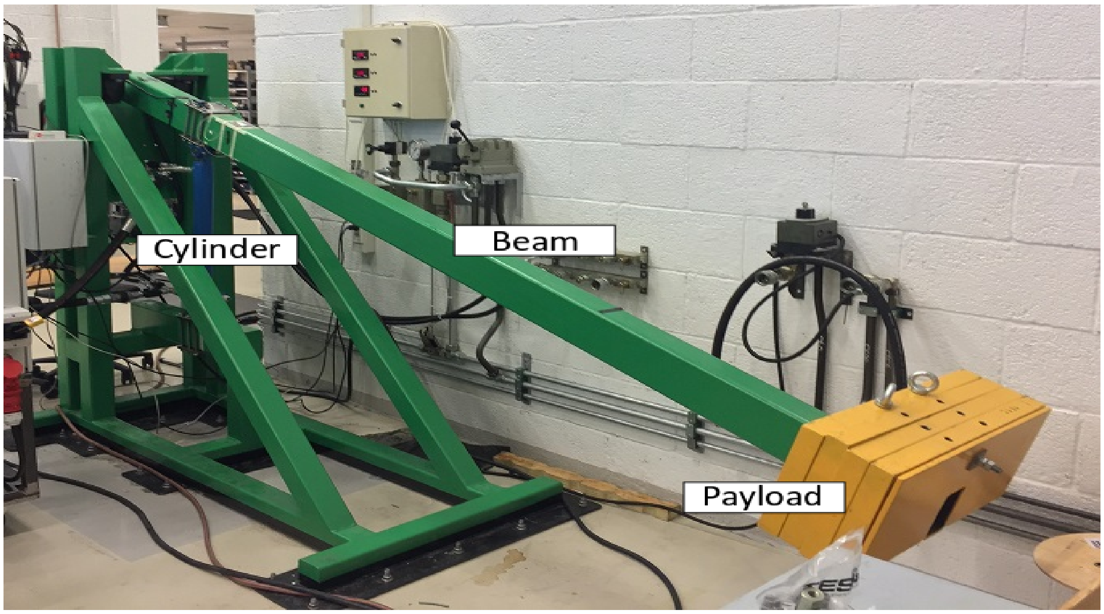


Figure 1: The single-boom crane structure

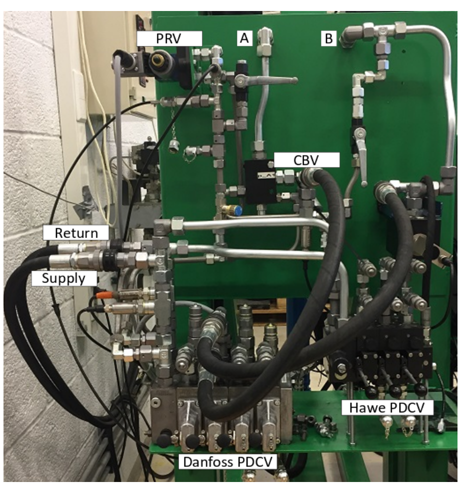


Figure 2: The original load-sensing system

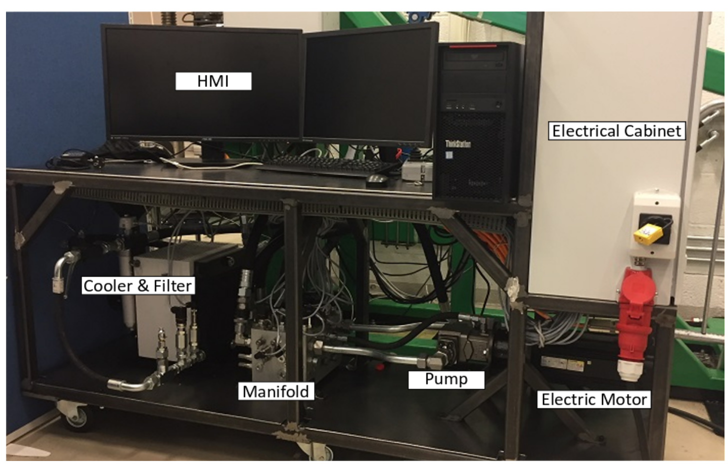


Figure 3: The self-contained system

1.2 Purpose of this manual

This manual aims to provide the user with the background necessary to operate the single-boom crane system using the two available methods: via the original central hydraulic system, and via the self-contained electro-hydraulic system. Both of these methods can be used manually, as well as electronically, for operation purposes. Additionally, this manual serves as a guide, when troubleshooting and/or modifying any of the systems with regards to their electrical and/or hydraulic subsystems.

1.3 Manual outline

This document is split up in many chapters, explaining the components used in the system, their installation recommendations and requirements, startup procedures, etc.

Single-Boom Crane contains the key mechanical characteristics of the common part for the two systems. This includes: the mechanical structure, the cylinder and switching valves.

Original System contains information about the original load-sensing system. The full hydraulic diagram, component list, and key components descriptions can be found here.

Self-Contained System contains information about ...

2 Single-Boom Crane

2.1 Structure

The single-boom crane beam is made of a rectangular, hollow steel beam, with steel plates attached as payload. The boom is attached to the main structure via a revolute joint at the end. The main structure is securely bolted down to the floor.

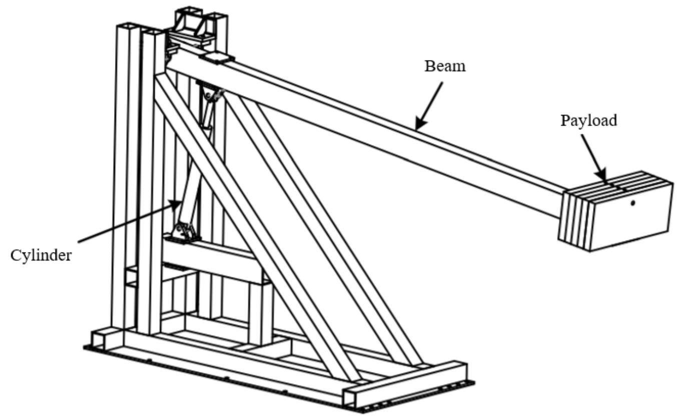


Figure 4: Single-boom crane structure drawing

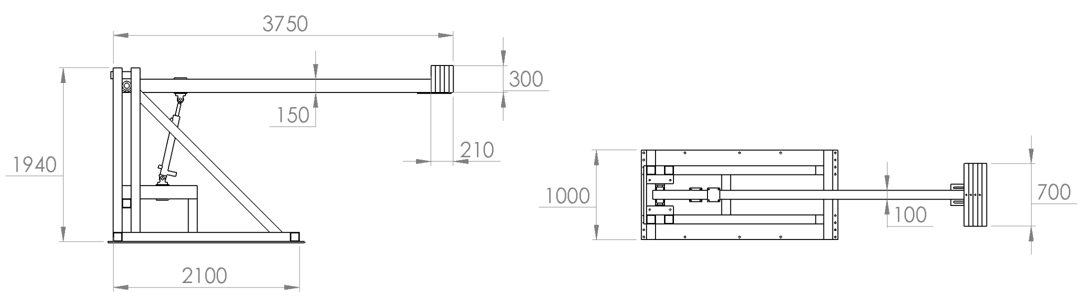


Figure 5: 2D Schematic of the single-boom crane structure

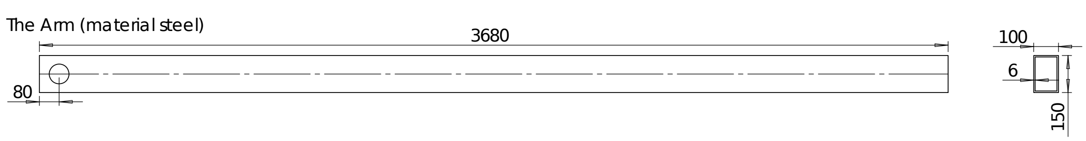


Figure 6: Beam and its cross-section dimensions

Parameter	Value	Unit
Density	7750-8050	kg/m^3
Payload plate mass	$4 \cdot 76$	kg
Payload end plate mass	16	kg
Payload bolt, nut, bracket mass	7.5	kg

Table 1: Single-boom crane parameters

During the motion of the crane, the equivalent mass pressing down on the cylinder, and the cylinder force required to satisfy force equilibrium are approximately:

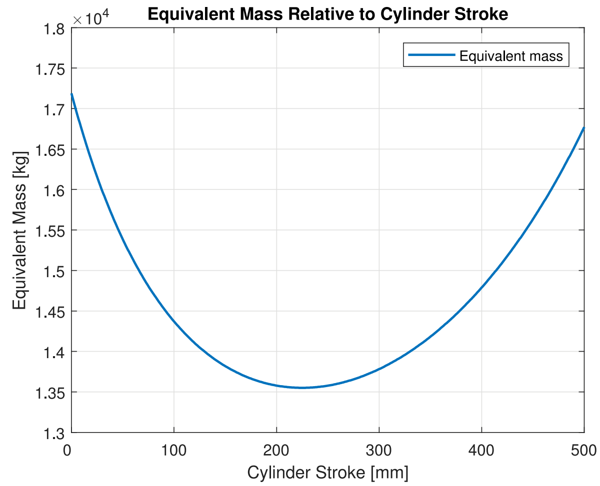


Figure 7: Equivalent mass acting on cylinder

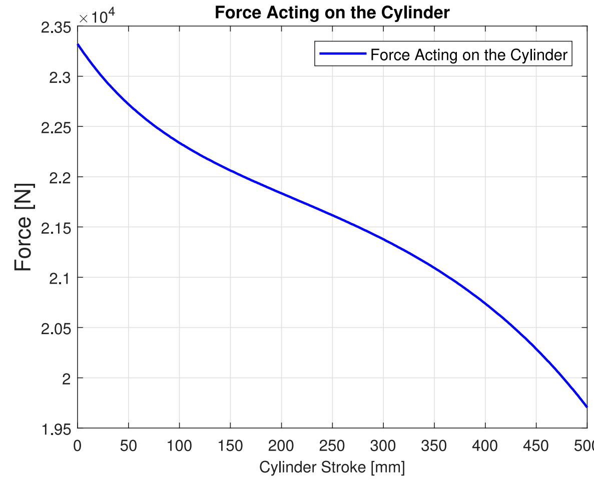


Figure 8: Equivalent force acting on cylinder

2.2 Cylinder

The hydraulic actuator attached to the crane base and the crane arm is a double-acting cylinder with a built in linear transducer which continuously reads the cylinder stroke. The cylinder is shown in Figure 10, as well as its key parameters in Table 2.

Parameter	Value	Unit
Cylinder diameter	65	mm
Piston rod diameter	35	mm
Stroke	500	mm
Mounting	Eye with spherical bearings	

Table 2: Cylinder parameters

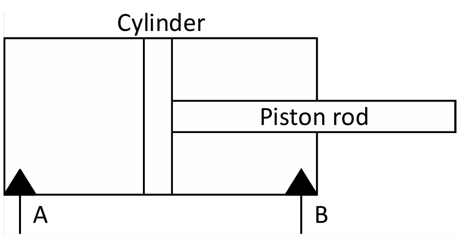


Figure 9: Cylinder diagram

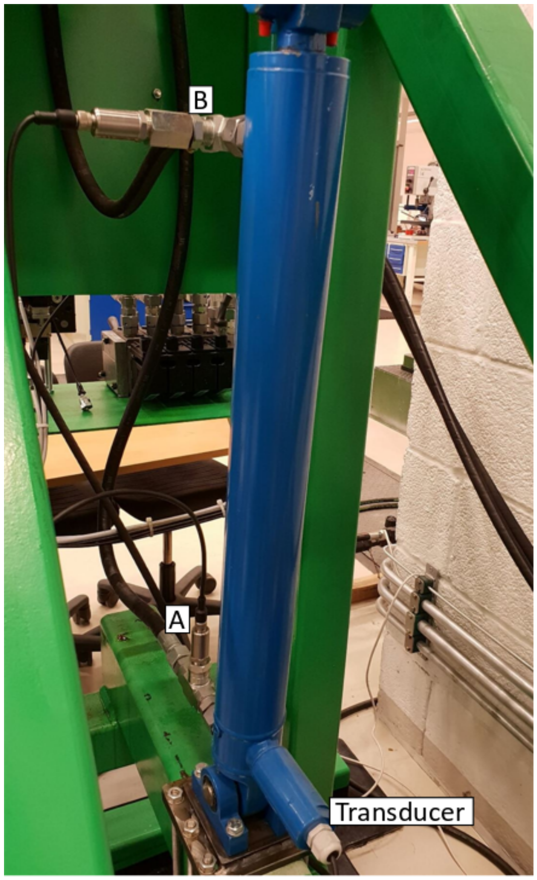


Figure 10: Hydraulic cylinder

2.3 Switch

Four on/off ball valves are mounted to the cylinder ports to allow the user switch between original load-sensing and self-contained systems with ease.

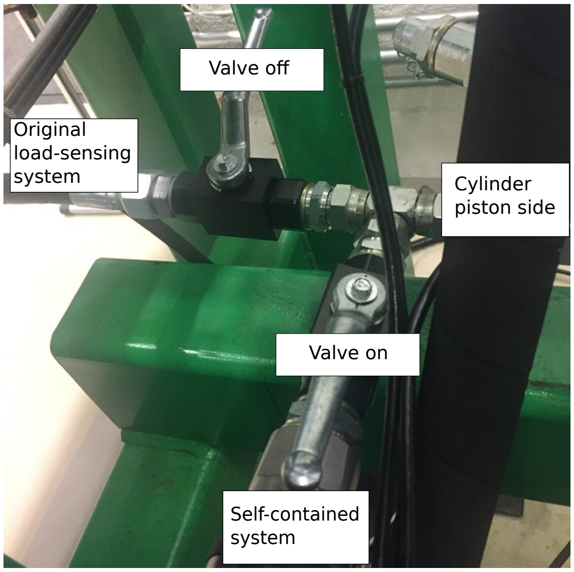


Figure 11: Cylinder piston side (A) switch valves

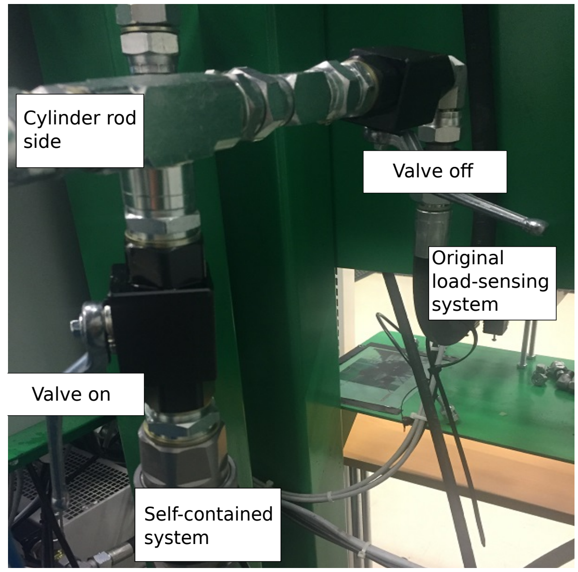


Figure 12: Cylinder rod side (B) switch valves

3 Original Load-Sensing System

3.1 Hydraulic Subsystem

Hydraulic diagram

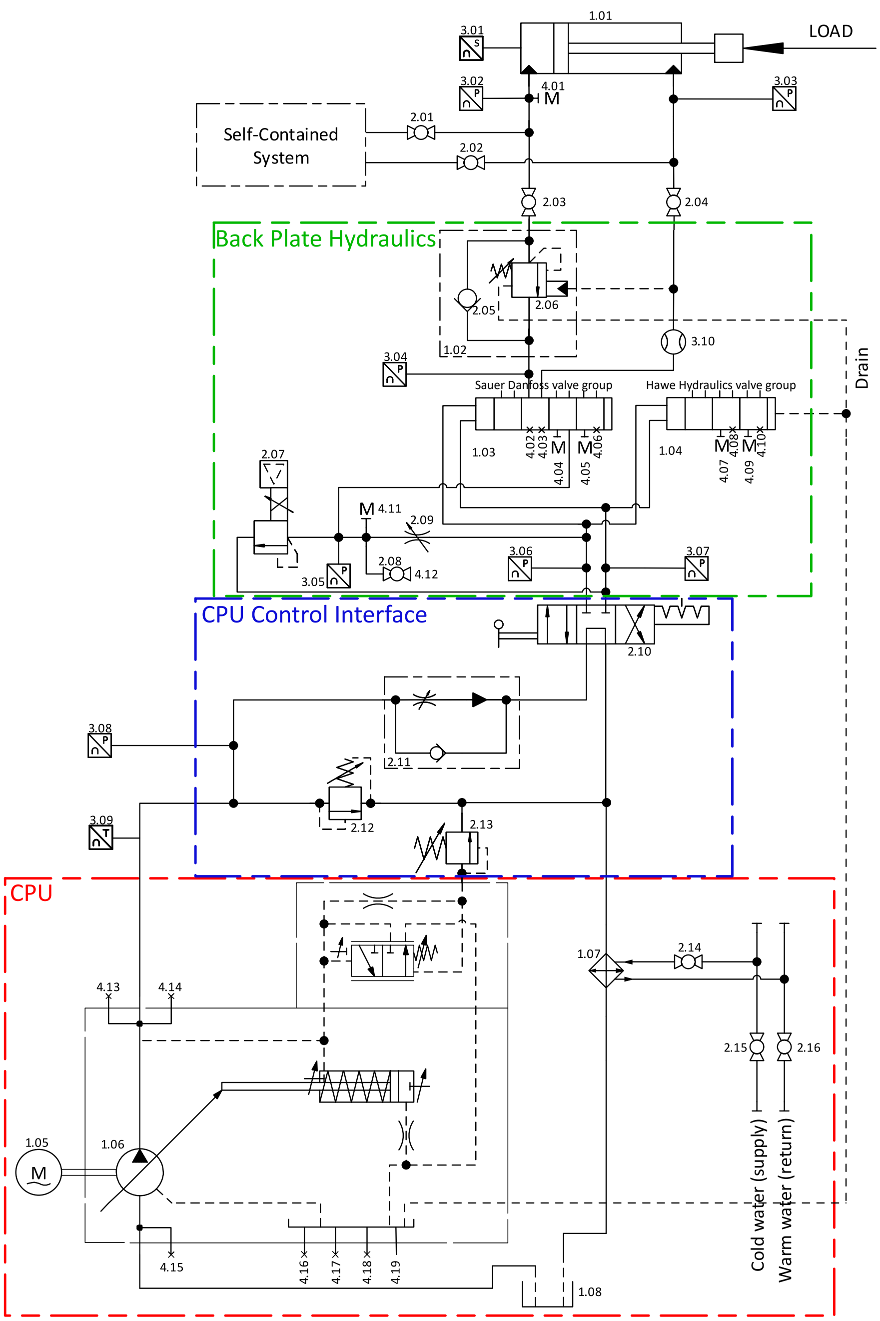


Figure 13: Original system hydraulic diagram (see additional diagrams for PVG and Hawe blocks below)

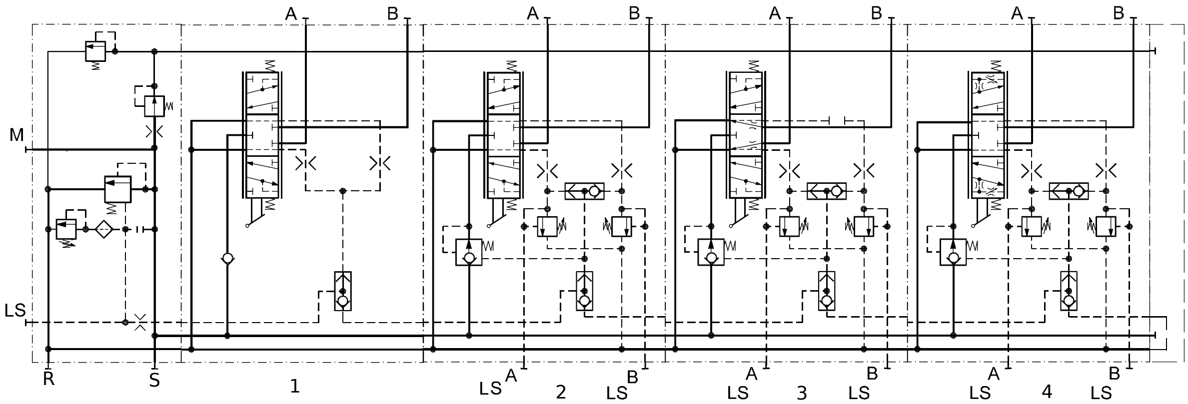


Figure 14: Proportional valve group from Sauer Danfoss (notice the modification on block 3)

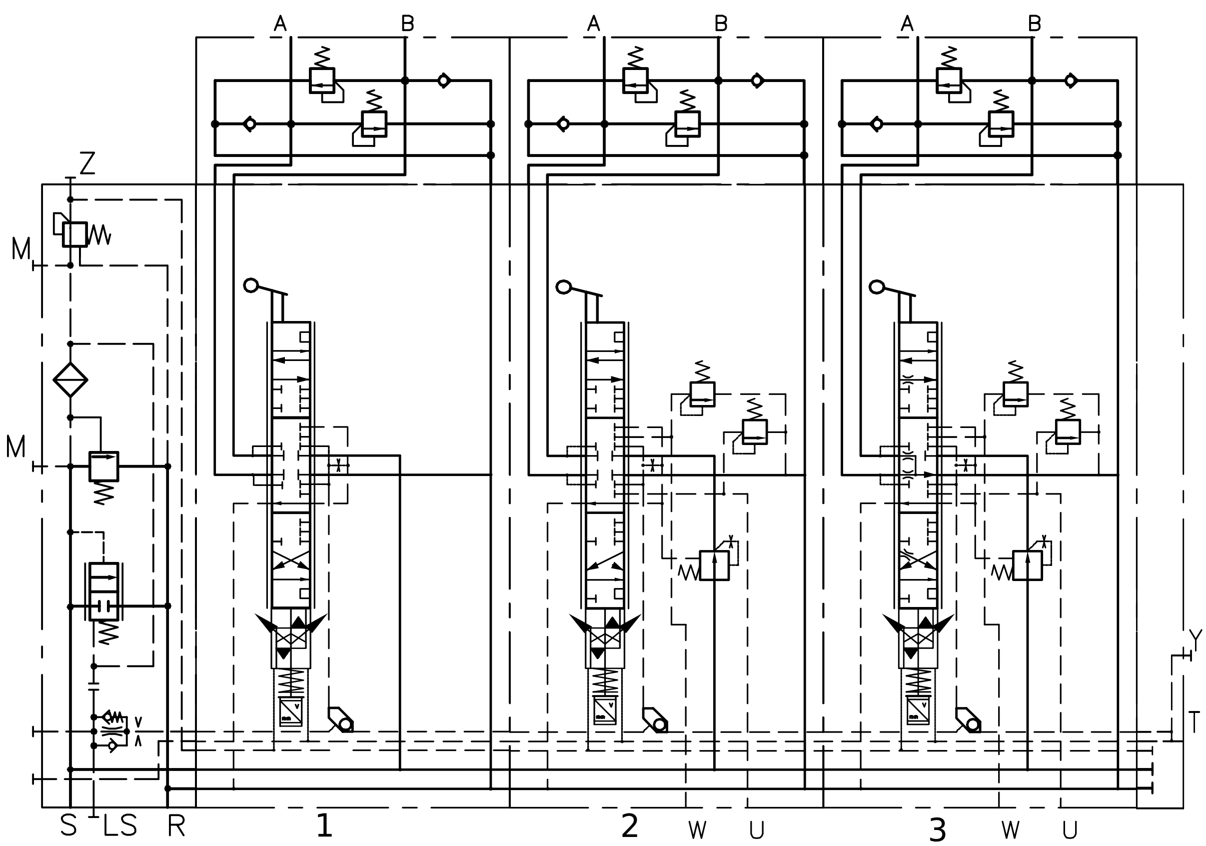


Figure 15: Proportional valve group from Hawe Hydraulics

Component list

Unit	Name	Manufacturer	Serial number
1.01	Hydraulic actuator	*PMC Cylinders*	$25CAL-65/35-0500/85$
1.02	Counterbalance valve	*SUN Hydraulics*	$CWCALHN-MAK$
1.03	Directional control valve group	*Sauer Danfoss*	custom
1.04	Directional control valve group	*Hawe Hydraulics*	specified in its own section
1.05	Electric motor	*ASEA*	$M225S60-4$
1.06	Hydraulic pump	*Brueninghaus Hydraulik*	$A4VS0\ 71\ DRG/10\ R-PPB\ 13\ NOO$
1.07	Cooler	-	-
1.08	Reservoir	-	-
2.01 - 2.04	Ball valve	-	-
2.05	Check valve	*SUN Hydraulics*	-
2.06	Pressure relief valve	*SUN Hydraulics*	-
2.07	Proportional pressure relief valve	*Bosch Rexroth*	$DBETE-6X/315\ G24\ K31\ A1\ V$
2.08	Ball valve	-	-
2.09	Flow control valve	-	$FT1251/2$
2.10	Directional valve	*Bosch Rexroth*	$H\ 4\ WMM\ 16\ G\ 30/F$
2.11	Flow control valve	*Bosch Rexroth*	$2\ FRM\ 16-32/100\ LB$
2.12	Pressure relief valve	*Bosch Rexroth*	$DBDH\ 6\ G17/400$
2.13	Pressure relief valve	*Bosch Rexroth*	$DBD\ 20\ G\ 11/$
2.14 - 2.16	Ball valve	-	-
3.01	Integrated position sensor	*PMC Cylinders*	-
3.02 - 3.08	Pressure sensor	*Parker*	$SCP-400$
3.09	Temperature sensor	-	-
3.10	Flow sensor	*Parker*	$SCQ-150$
4.01 - 4.19	Mesh point	-	-

Table 3: Component table for the original load-sensing system

Central power unit

The central power unit (Figure 16) that supplies the original load-sensing system consists of the electric motor, the variable displacement pump and the reservoir and is located in the hydraulic room nearby the crane. The control panel (Figure 17) is mounted on the wall next to the crane. The Supply* and Return* hoses from Figure 17 connect to the Supply and Return in the original load-sensing system in Figure 2.

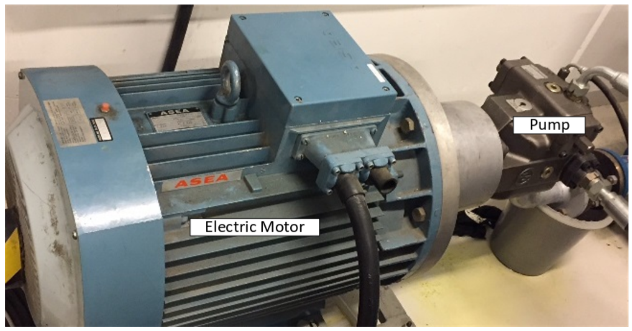


Figure 16: Central power unit

Parameter	Value	Unit
Maximum pressure	200	*bar*
Maximum flow	100	l/min

Table 4: Central power unit parameters

Central power unit controls

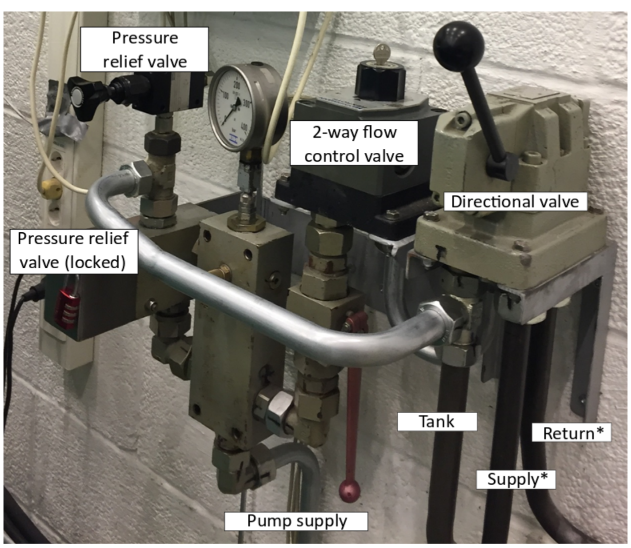


Figure 17: CPU controls

Figure 18: Emergency stop

- Water cooling to the CPU is turned on from the hydraulic room next to the crane.
- To start the system, release the emergency stop and press the ”up” arrow button.
- Set the desired supply pressure using the turn knob on the pressure relief valve in Figure 17.
- Set the desired flow using the turn knob on the 2-way flow control valve in Figure 17.

Important: Before starting the system, make sure water cooling is on, supply pressure and flow are set to 0 (manometer shows around 24bar at lowest setting). Once the system is running, set the supply pressure before setting the flow. When shutting the system off, set the flow to 0, then set supply pressure to 0, before pressing the emergency stop button.

Proportional directional control valves

The original load-sensing system has two proportional directional control valve block groups the user can connect to. The first block group is from *Sauer Danfoss* and the second block group is from *Hawe Hydraulics*.

Sauer Danfoss valve group

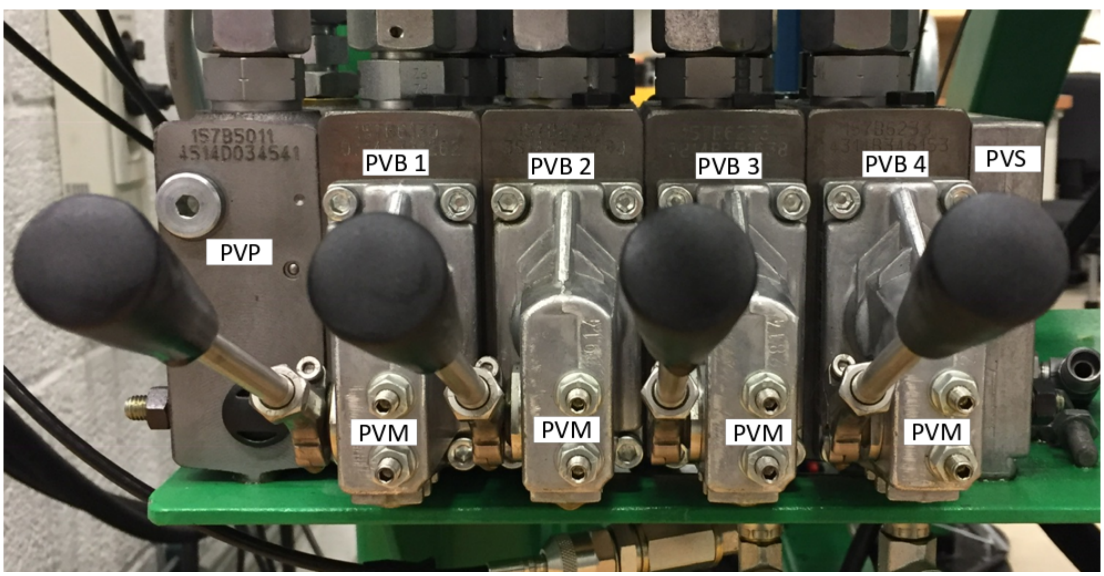


Figure 19: Front view of *Sauer Danfoss* valve group (*PVES-SP modules behind not visible)

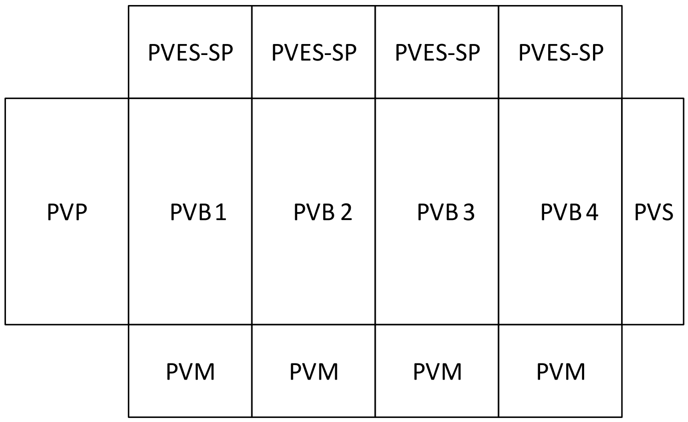


Figure 20: Top-down layout of *Sauer Danfoss* valve group

Annotation	Block name
PVP	Pump side module
PVB	Basic module
PVS	End plate
PVES-SP	Electric actuation module
PVM	Mechanical actuation module

Table 5: Annotation meanings for Figure 20

PVB 1	PVB 2	PVB 3	PVB 4
Load drop check valve	Non-damped compensator valve	Non-damped compensator valve	Non-damped compensator valve
Closed neutral position	Closed neutral position	Open neutral position	Closed neutral position
Flow control spool with linear flow characteristic	Flow control spool with linear flow characteristic	Flow control spool with linear flow characteristic	Standard pressure control spool
Without load-sensing shuttle valve	With load-sensing shuttle valve, $LS_{A/B} = 200bar$ (max)	With load-sensing shuttle valve, $LS_{A/B} = 200bar$ (max)	With load-sensing shuttle valve, $LS_{A/B} = 200bar$ (max)
Mechanical and electrical actuation, with shock and anti-cavitation valves (setting: $265bar$)			

Table 6: *Sauer Danfoss* valve group characteristics

Hawe Hydraulics valve group

Serial no:
$PSV\ 31/D250 - 2$
$-A\ 1\ L\ 25/16/EAWA/2\ AN\ 265\ BN\ 265$
$-A\ 2\ L\ 25/16\ A\ 200\ B\ 200S\ 1/EAWA/2\ AN\ 265\ BN\ 265$
$-A\ 2\ O\ 25/16\ A\ 200\ B\ 200S\ 1/EAWA/2\ AN\ 265\ BN\ 265$
$-E\ 1 - G\ 24$

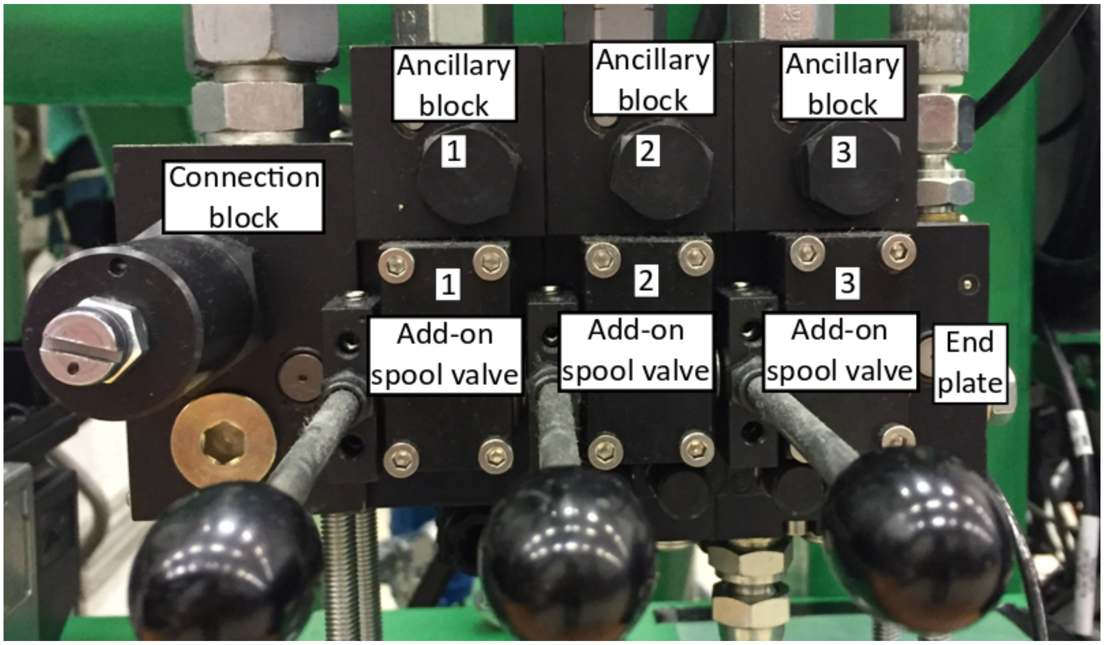


Figure 21: Front view of *Hawe Hydraulics* valve group

Counterbalance valve

The counterbalance valve is a safety valve installed for the purpose of passive load holding and leak prevention. The valve sits inside the manifold shown in Figure 22. Its parameters are listed in Table 8.

Parameter	Value	Unit
Pilot ratio	3 : 1	–
Set pressure	196	*bar*

Table 8: Over-centre valve parameters

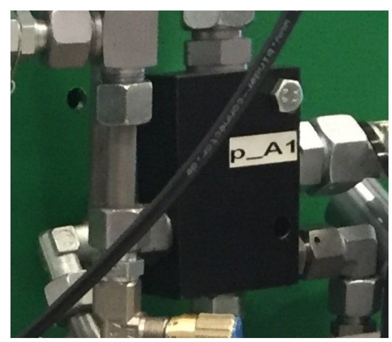


Figure 22: Manifold for the over-centre valve

Part	1	2	3
Connection block	Made for variable displacement pumps, with integrated pressure reducing valve for internal supply of control oil (set to $20bar$) and with pressure limiting valve set to $250bar$		
Add-on spool valve	Closed neutral position, nominal max flow in A and B rated at $dp = 5bar$ of 25 and $16L/min$, electro-hydraulic and manual actuation with spool position feedback		A and B connected to tank via throttle in neutral position, nominal max flow in A and B rated at $dp = 5bar$ of 25 and $16L/min$, electro-hydraulic and manual actuation with spool position feedback
	No compensator, no functional cut-off	Pressure compensated, LS_A and LS_B set to $200bar$, functional cut-off plus proportional pressure compensation	
Ancillary block	Shock and suction valves in A and B ports set to $265bar$		
End plate	Standard end plate, $12V$ electrical actuation		

Table 7: *Hawe Hydraulics* valve group characteristics

When the oil flows upwards (from A1 to A), it passes the bypass check valve on the left. When the oil flows downwards (from A to A1), the check valve on the left is blocked and pressures A and B build up until they open the over-centre valve. The pilot ratio 3:1 means that pressure B is acting on an area 3 times larger than pressure A.

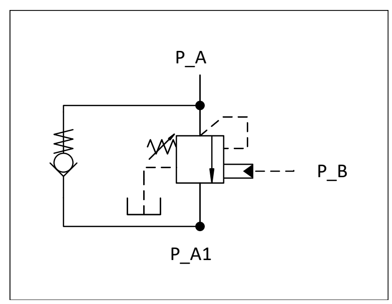


Figure 23: Hydraulic diagram of the over-centre valve

3.2 Electrical Subsystem

Sauer Danfoss valve group electrical actuation

The Sauer Danfoss directional control valve group has electrical actuation modules. Signal wire connections from the module in Figure 24 are described in Table 9.

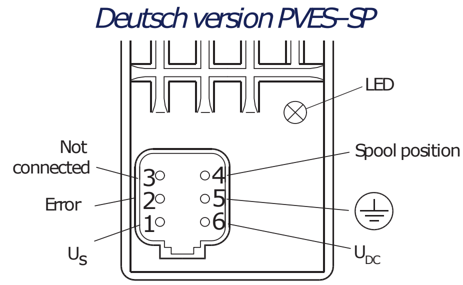


Figure 24: PVES-SP module connections (source: Sauer Danfoss)

Port	In/Out	Range	Information
1. U_S	In	$(15\% - 85\%) \cdot U_{DC}$	Control voltage (DC)
2. Error	Out	$100\ mA$ max	Error signal
3. Not connected	-	-	-
4. Spool position	Out	$(0.5 - 4.5)V$	Position feedback signal
5. Ground	-	-	Ground
6. U_{DC}	In	$(11 - 32)V$	Supply voltage (DC)

Table 9: PVES-SP connection signals

Hawe Hydraulics valve group electrical actuation

The Hawe Hydraulics directional control valve group has electrical actuation modules. Signal wire connections from the module in Figure 25 are described in Table 10.

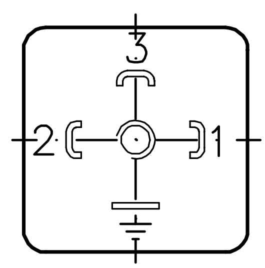


Figure 25: G 24 module connections (source: Hawe Hydraulics)

Port	In/Out	Range	Information
1. U_{OUT}	Out	-	Output voltage (DC)
2. U_B	In	$(5 - 10)V$	Operating voltage (DC)
3. Ground	-	-	Ground

Table 10: G 24 connection signals

4 Self-Contained System

General introduction to self contained systems

4.1 Hydraulic subsystem

Hydraulic pump connections

describe pump-motor mounting
The hydraulic pump and the electric motor is connected with a xxxx. this connection is located inside the bell housing. The motor and pump is bolted to the bell housing, which in turn is bolted to the motor mount shown in the figure below:

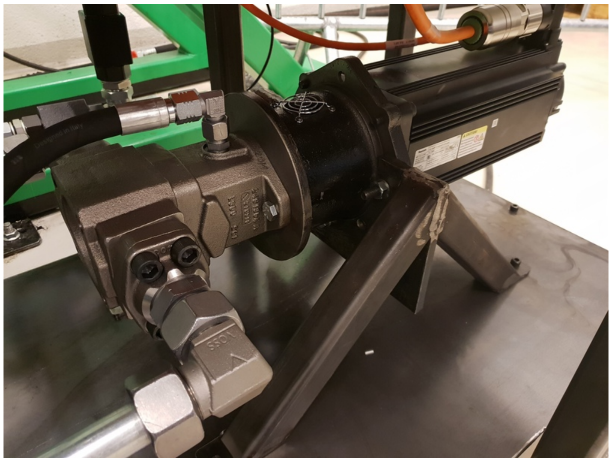


Figure 26: Piture of motor mount, electric motor, and pump

The connection between the pump and the manifold can be done with hoses or pipes. The 1 inch diameter is kept throughout the pup-manifold connection.

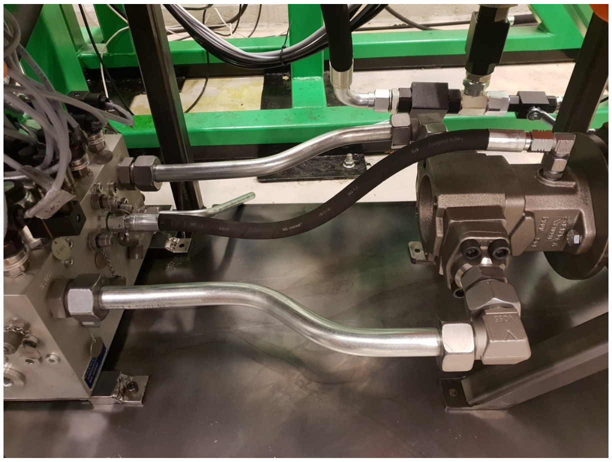


Figure 27: Piture of motor mount, electric motor, and pump

Other technical data is shown in the table below:

Ports	Manifold	Pump
Piston side	5.002	B
Rod side	5.001	A

Table 11: Parameters for electric motor and pump

Parameter	Value	Unit
Maximum pressure	350	bar
Maximum flow	1??	l/min

Table 12: Parameters for electric motor and pump

Oil filter and cooler connections

The cooler used in the new system is an oil/air cooler from Bosch Rexroth, ordering code KOL3N-30/R-F100-10-O/M. It has a cooling power of $3kW$ and supply voltage $230/400V$ at $50Hz$. In addition, the cooler has a hydraulic oil filter mounted. The filter is also from Bosch Rexroth, ordering code 50LEN0100-H10XLA00-V2.2-M-R3. It allows a maximum filter flow of $75l/min$.
The filter is is bolted directly on the cooler unit. It is KOL3N-30/R-F100-10-O/M as can bee seen in the figure below (insert new picture here).
The ports xx go from the manifold to the filter intake.
There are also two temperature sensors, one pressure sensor and one mini mess connector on the cooler unit. They can be seen in figure xx below as well as marked out in the hydraulic schematic to the right.
Two ball valves and a t- connection has been added to enable the operator to bypass the cooler unit. They can be seen in figure xx below as well as marked out in the hydraulic schematic to the right.
The ports xx go from the cooler out port to the manifold opening xx.
Other technical data is shown in the table below:

Actuator connections

describe connections from actuator to manifold
The actuator is connected to the manifold trough hoses, quick couplings, ball valves and t-connections. They can be seen in figure xx below as well as marked out in the hydraulic schematic to the right.

Figure 28: The manifold placed on the trolley.

Other technical data is shown in the table below:

Parameter	Value	Unit
Maximum pressure	200	bar
Maximum flow	100	l/min

Table 13: Central power unit parameters

Manifold connections

describe remaining manifold connections
describe sensor connections to the manifold
The hydraulic manifold block used for the new system is costume made by Hydman, and includes multiple different components. All the components are listed in table 14

Component	Code	Type	Quantity
Pilot to open check valve	CHESU02001	CKEB XCN	2
Pilot to open check valve	CHESU22001	CVEV XFN	2
Check valve	CHEHA04001	RK4	3
Check valve	CHEHA02000	RB2	3
Check valve	CHEHA02001	RK2	2
Pressure relief valve	RELSU10019	RDDA LAN + CAPNUT 400158	4
Pressure relief valve	RELSU10041	RDDA LSN + HAND KNOB	1
Orifice	KUR0080080	M8-0.8	1
Measuring connector	MITST14000	SMK20-G1/4-PC	8
Pressure sensor	-	Rexroth HM20-21/10-C-K35	2
Pressure sensor	-	Rexroth HM20-11/250-C-K35	8
Temperature sensor 4-20 mA	STDA000008	MBT 3560-0000-0050-10-110 084Z403	10
3/2-directional poppet valve	DIRAH34015	SD1E-A3/H2S8M9	1
Usit seal	US00400000	U/SC 13,74x20,57x2,0 G1/4	10
Coil	KEAH000005	C19B-02400E1-25,75NA 27670600	1
Hydraulic system manifold	HM00027500	000275	1
Plug	TUL0180000	G 1/8" A	18
Plug	TUL0140000	G 1/4" A	2
Plug	TUL0380000	G 3/8" A	6
Plug	TUL0120000	G 1/2" A	10
Plug	TUL1000000	G 1" A	3
Lifting eye bolt	NO08000000	M8 DIN580 ST	2
Label	HM40016000	HYDMAN 30x60	1

Table 14: Manifold components.

The RDDA LAN pressure relief valve has a cracking pressure of $200bar$, whereas cracking pressure for the RDDA LSN relief valve is $14bar$. The 3/2 directional control valve has a maximum operating pressure of $350bar$ and a maximum flow setting of $25l/min$.

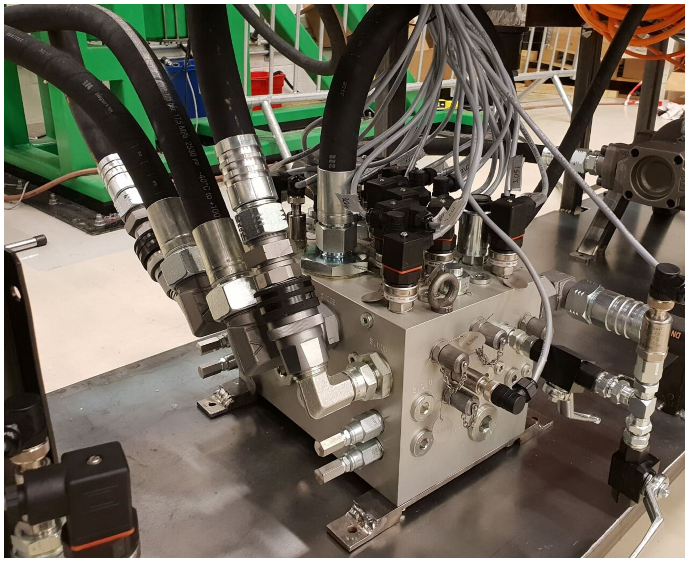


Figure 29: The manifold placed on the trolley.

As can be seen in table 14, the manifold has several sensors attached to it. The two kinds of pressure sensors are from Bosch Rexroth and can respectively measure pressure up to 250 and $10bar$. Both kinds has a current output signal of 4 to $20mA$. The temperature sensors is from Sauer Danfoss, and has a measuring range from -50 to $200°C$. Like for the pressure sensors, they have a current output signal of 4 to $20mA$.

Accumulator connections

The accumulator is a $10l$ bladder-type accumulator from Bosch Rexroth, ordering number HAB10-330-60/0G09G-2N111-CE. The maximum operating pressure in the accumulator is $330bar$ and the maximum flow is $900l/min$.
describe connections from accumulator to manifold
The accumulator is connected to the manifold trough hoses, ball valves and t-connections. They can be seen in figure xx below as well as marked out in the hydraulic schematic to the right.

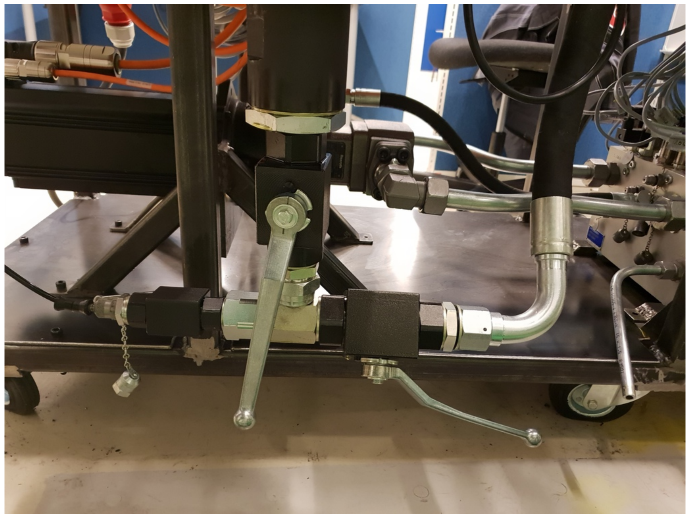


Figure 30: The accumulator connection

Other technical data is shown in the table below:

Parameter	Value	Unit
Maximum pressure	350	bar
Maximum volume	$9,2$	l
Pre-pressure	1,2	bar

Table 15: Central power unit parameters

System priming

describe priming of system with oil and airing process
and components involved in the process

4.2 Electronics

Electric motor

The electric motor used in the new system is a permanent-magnet synchronous motor (PMSM) from Bosch Rexroth, ordering code MSK071E-0300-NN-S1-UG0-NNNN0. It has a maximum torque of $84Nm$ and a maximum rotational speed of $4200rpm$. Furthermore, the motor has a cooling system of natural convection included.

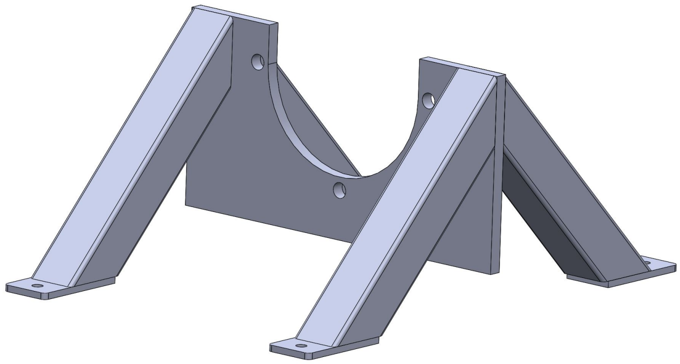


Figure 31: PMSM motor.

requirements, mounting, connections

Motor drive

The motor drive is an IndraDrive C HCS02.1E-W0054-A-03-NNNN drive from Bosch Rexroth. The drive has a maximum current of $54A$ and a maximum output of $10kW$ with choke.
motor drive installation requirements
connections to PMSM (motor)
connections to other components

Safety switches and circuit breakers

The safety relay utilised in the cabinet is a electromechnical relay from Phoenix Contact, ordering code PSR-SCP-24DC/ESP4/2X1/1X2. The relay is made for use for both high- and low demand applications. It has a supply voltage of $24V$ DC and a maximum switching voltage at the output of $250V$ AC/DC.

Figure 32: The safety relay.

describe the safety switch and circuit breakers used and their connections to the overall circuit

Sensors

The manifold has several sensors attached to it. The two kinds of pressure sensors are from Bosch Rexroth and can respectively measure pressure up to 250 and $10 bar$. Both kinds has a current output signal of 4 to $20 mA$. The temperature sensors is from Sauer Danfoss, and has a measuring range from -50 to $200°C$. Like for the pressure sensors, they have a current output signal of 4 to $20 mA$. The sensors are shown in the figures below.

(a) Danfoss temperature sensor.

(b) Rexroth pressure sensor.

Figure 33: Sensors on the manifold.

describe all sensor connections in the cabinet

Programmable logic controller

The programmable logic controller (PLC) utilised in the self-contained cylinder consists of one central processing unit (CPU), three AI-modules, one AI/AO-module, one DI-module and one DO-module, all from Bosch Rexroth. The CPU is an IndraControl XM22. The three AI-modules are of type S20-AI-8 and consist of eight analog inputs each. The AI/AO-module is of type S20-AI6-AO2-SSI2 and consists of six analog inputs, two analog outputs and two SSI interfaces. The digital input is of type S20-DI-16/4, and consist of 16 digital inputs. The digital output is of type S20-DO-16/3, and consists of 16 digital outputs.

Figure 34: The PLC mounted on the cabinet.

plc requirements
connections to other comps.

Line filter

line filter reqs.
connections to other comps.

Brake resistor

The brake resistor is from Bosch Rexroth, type code HLR01.1N-03K8-N40R3. The brake resistor has a continuous power rating of $4kW$ and a break resistance of $40.3Ohm$.
requirements and connections

Current converters

The AC/DC converter is from Mean Well, model type NDR-480-24, and is a part of the DIN rail power supply series. It includes a full range universal AC input, and has a output voltage of $24V$ and a output power of $480W$. requirements and connections
requirements and connections

Wire terminals

Wiring

overall wiring in the cabinet
Drawings inn here

Appendices

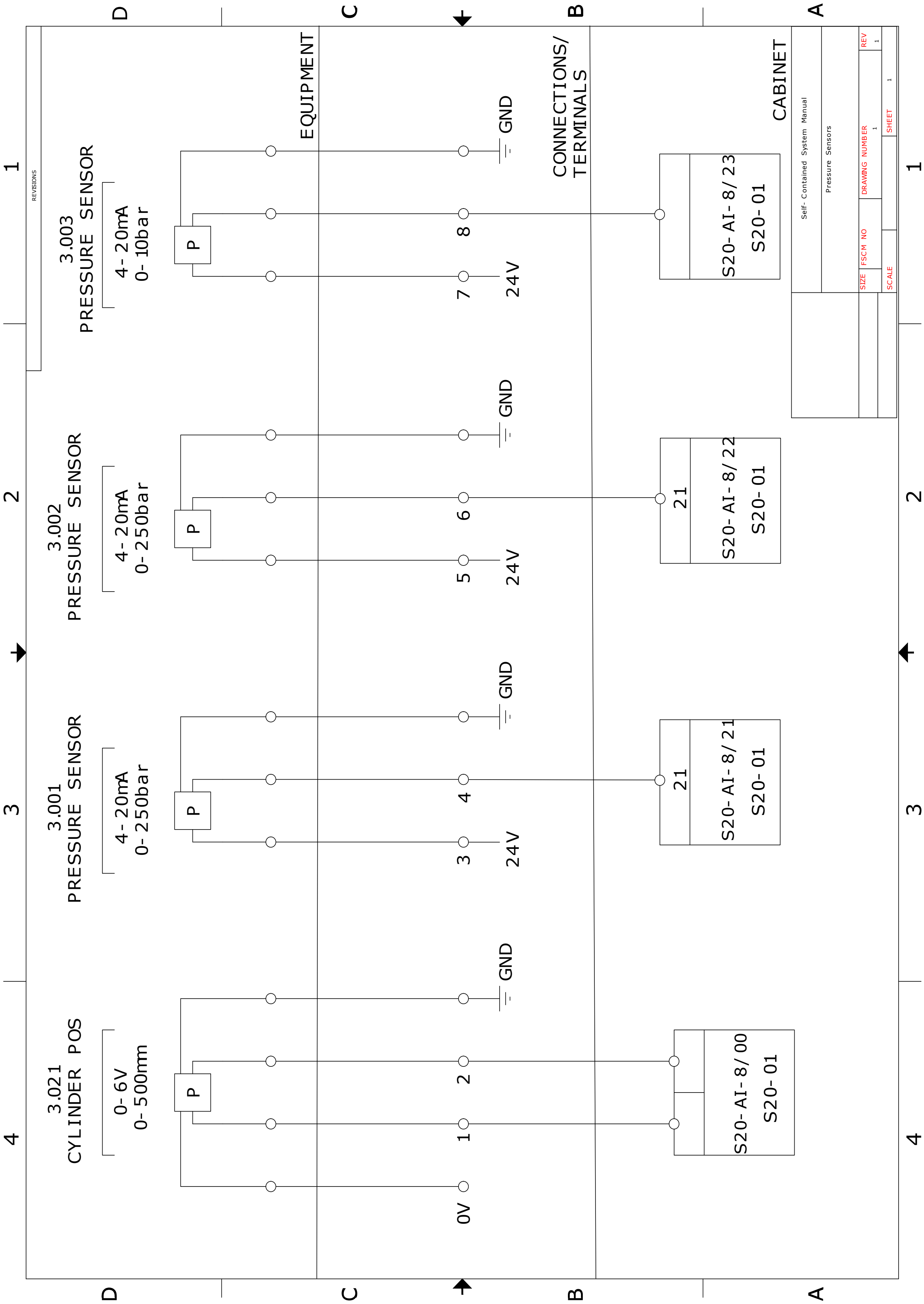
Hydraulic schematics

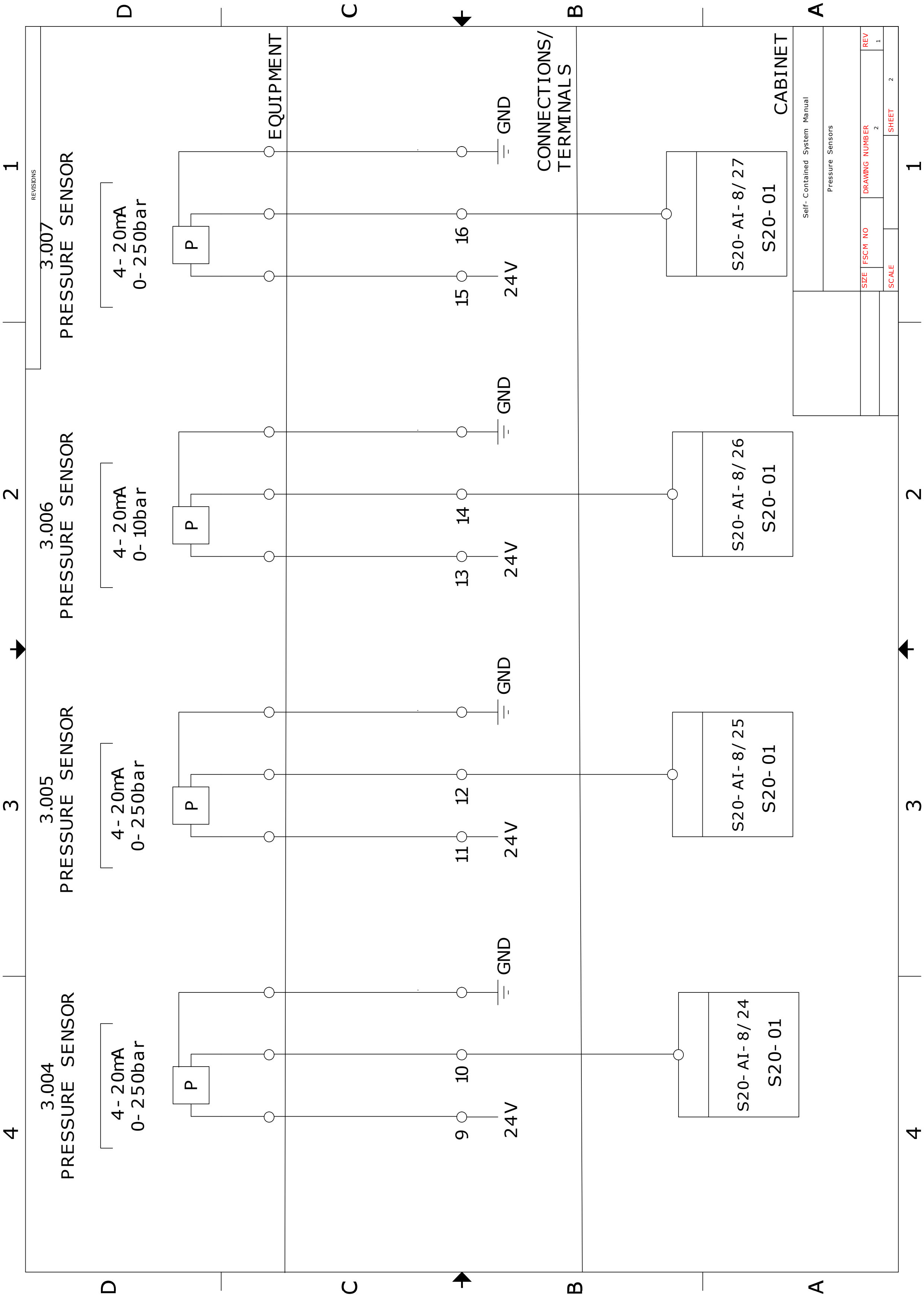
self-explained

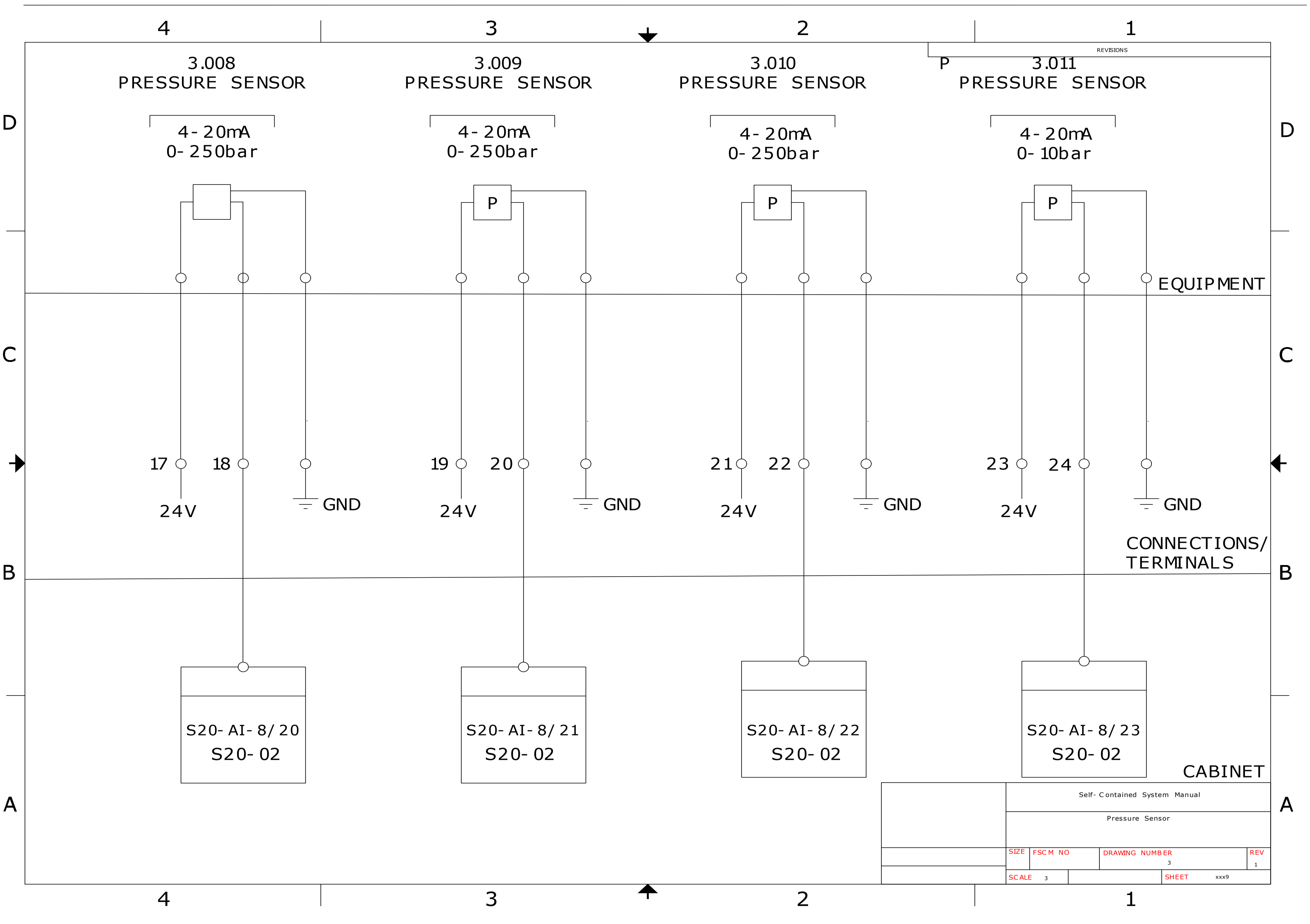
Electrical schematics

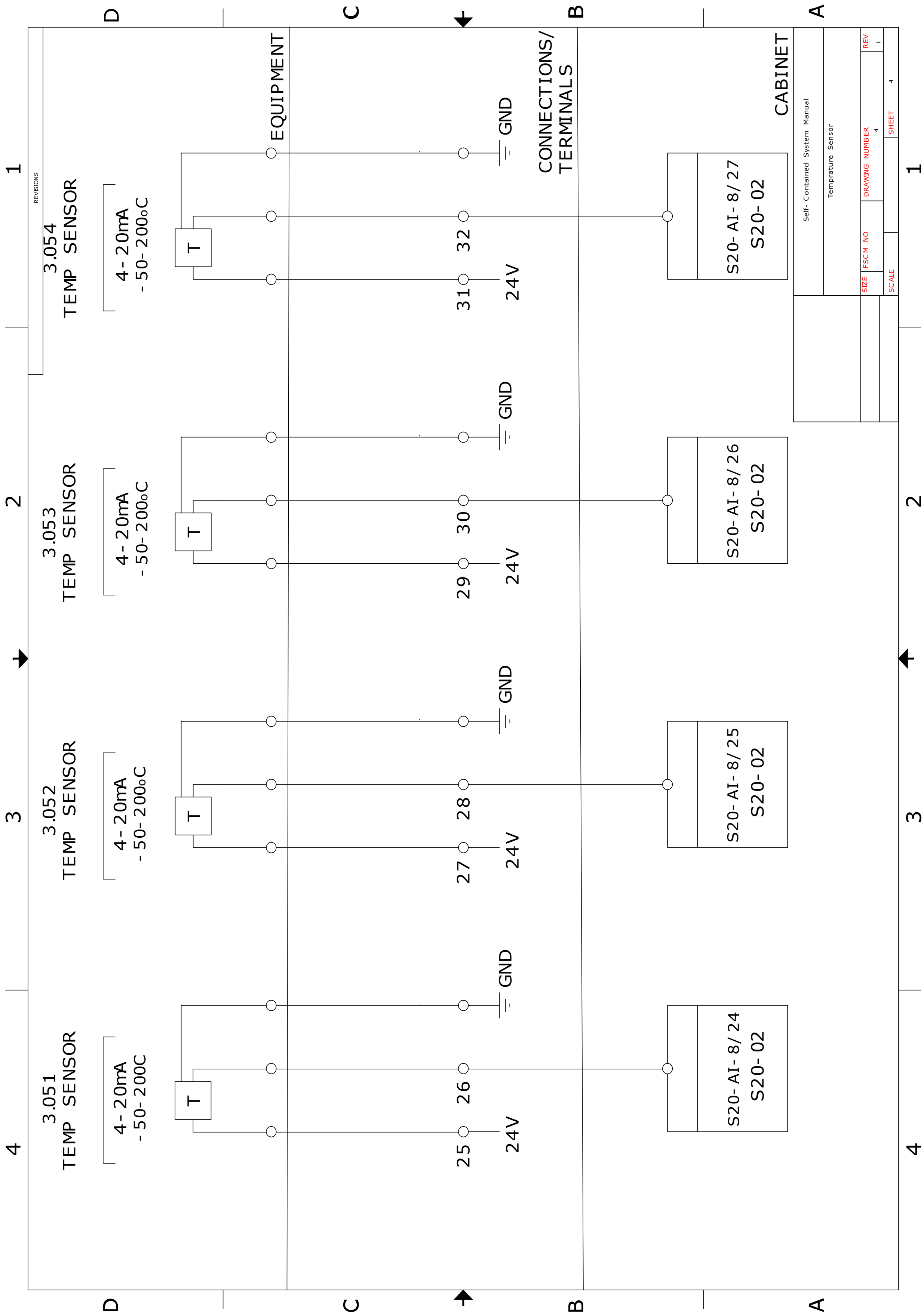
Drawings

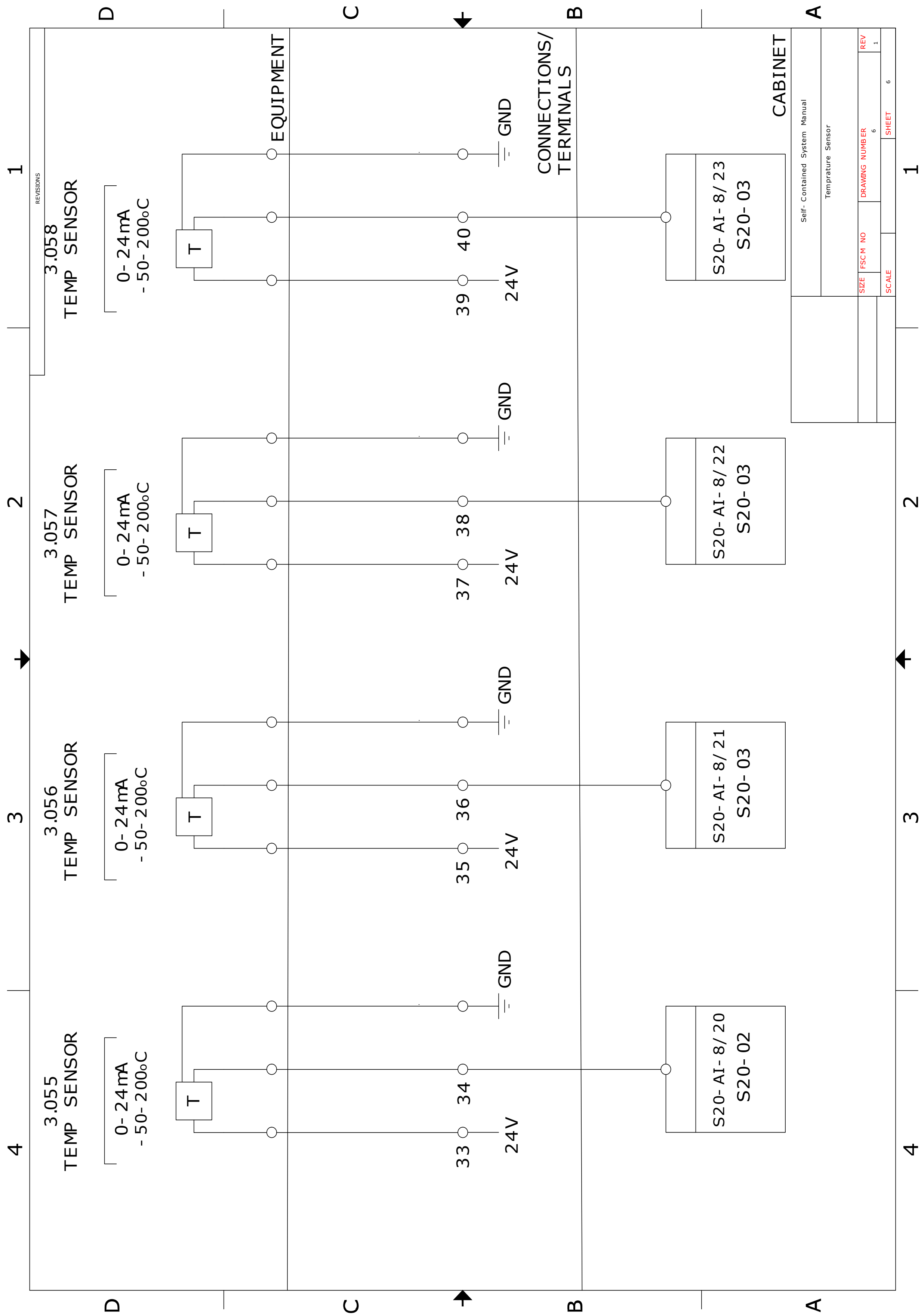
CAD drawings of components?

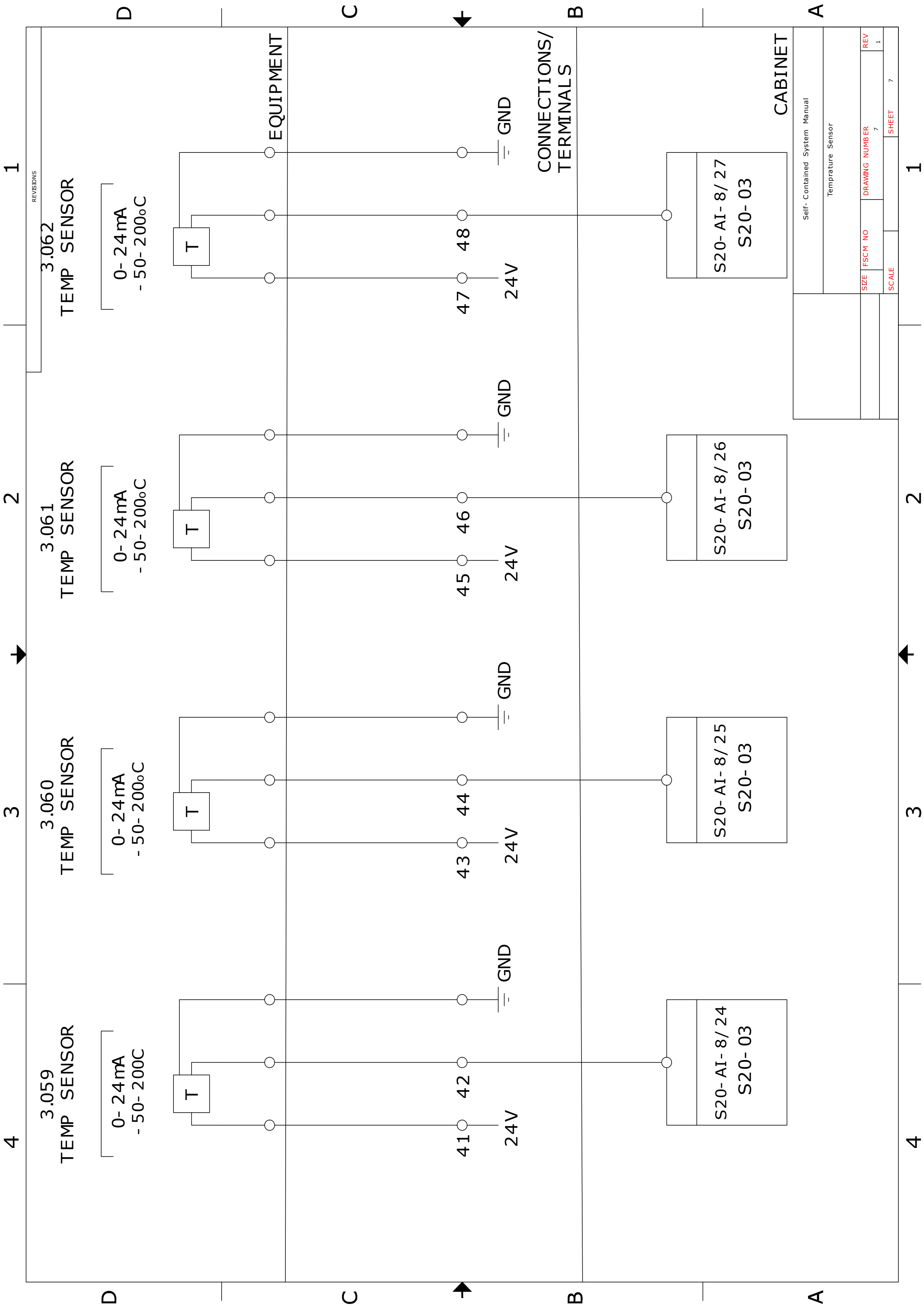

REVISIONS
3.021
CYLINDER POS
0-6V
0-500mm
3.001
PRESSURE SENSOR
4-20mA
0-250bar
3.002
PRESSURE SENSOR
4-20mA
0-250bar
3.003
PRESSURE SENSOR
4-20mA
0-10bar
P
EQUIPMENT
0V
1
2
3
4
5
6
7
8
24V
GND
CONNECTIONS/
TERMINALS
21
S20-AI-8/00
S20-01
S20-AI-8/21
S20-01
S20-AI-8/22
S20-01
S20-AI-8/23
S20-01
CABINET
Self-Contained System Manual
Pressure Sensors
SIZE
FSCM NO
DRAWING NUMBER
REV
SCALE
SHEET
1
A
B
C
D


REVISIONS
3.007
PRESSURE SENSOR
4-20mA
0-250bar
3.006
PRESSURE SENSOR
4-20mA
0-10bar
3.005
PRESSURE SENSOR
4-20mA
0-250bar
3.004
PRESSURE SENSOR
4-20mA
0-250bar
P
EQUIPMENT
GND
24V
9
10
11
12
13
14
15
16
CONNECTIONS/
TERMINALS
CABINET
S20-AI-8/27
S20-01
S20-AI-8/26
S20-01
S20-AI-8/25
S20-01
S20-AI-8/24
S20-01
Self-Contained System Manual
Pressure Sensors
SIZE
FSCM NO
DRAWING NUMBER
2
REV
1
SCALE
SHEET
2

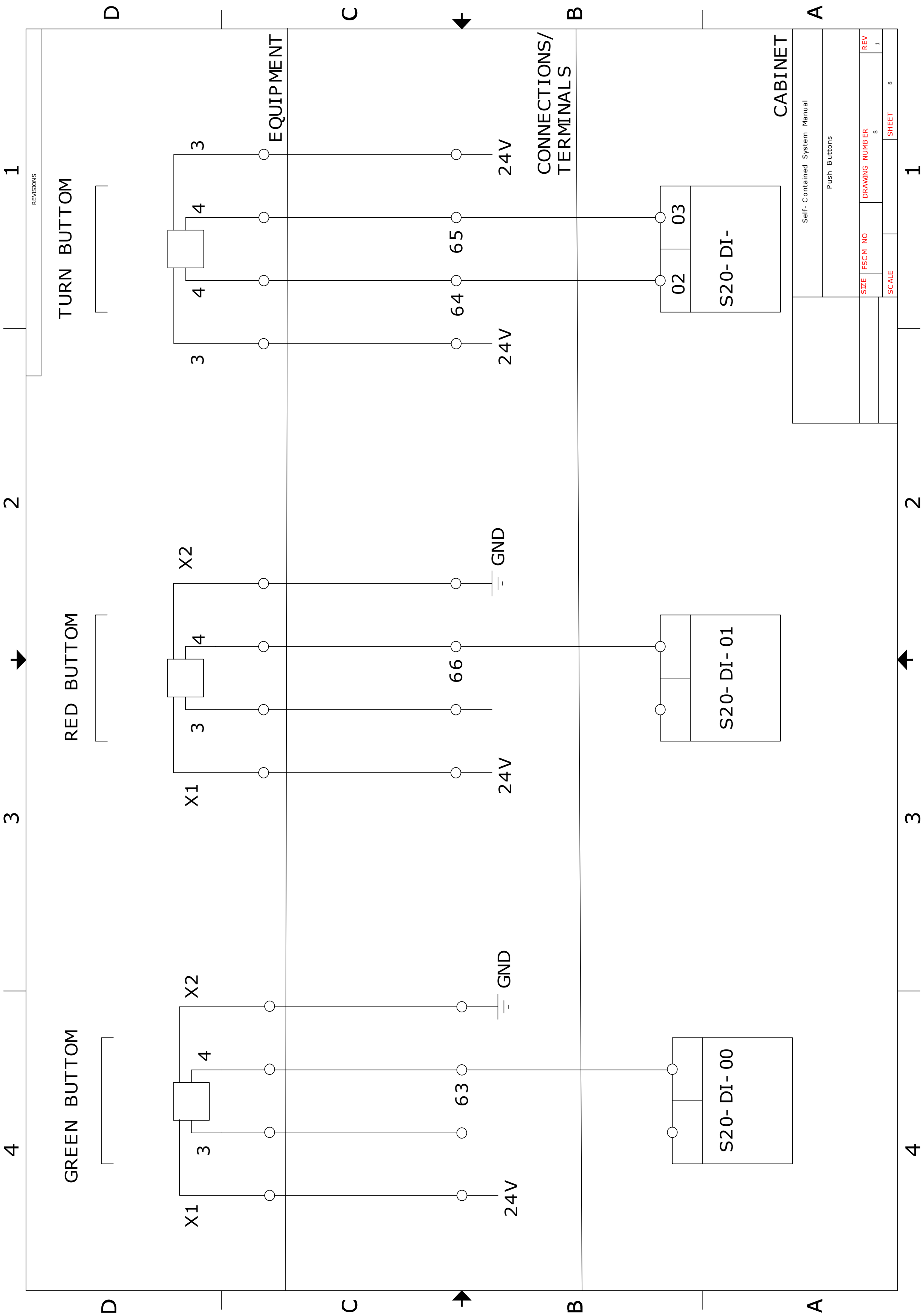

3.008
PRESSURE SENSOR
4-20mA
0-250bar
3.009
PRESSURE SENSOR
4-20mA
0-250bar
3.010
PRESSURE SENSOR
4-20mA
0-250bar
3.011
PRESSURE SENSOR
4-20mA
0-10bar
P
EQUIPMENT
17
18
19
20
21
22
23
24
24V
GND
CONNECTIONS/
TERMINALS
S20-AI-8/20
S20-02
S20-AI-8/21
S20-02
S20-AI-8/22
S20-02
S20-AI-8/23
S20-02
CABINET
REVISIONS
Self-Contained System Manual
Pressure Sensor
SIZE
FSCM NO
DRAWING NUMBER
3
REV
1
SCALE
3
SHEET
xxx9

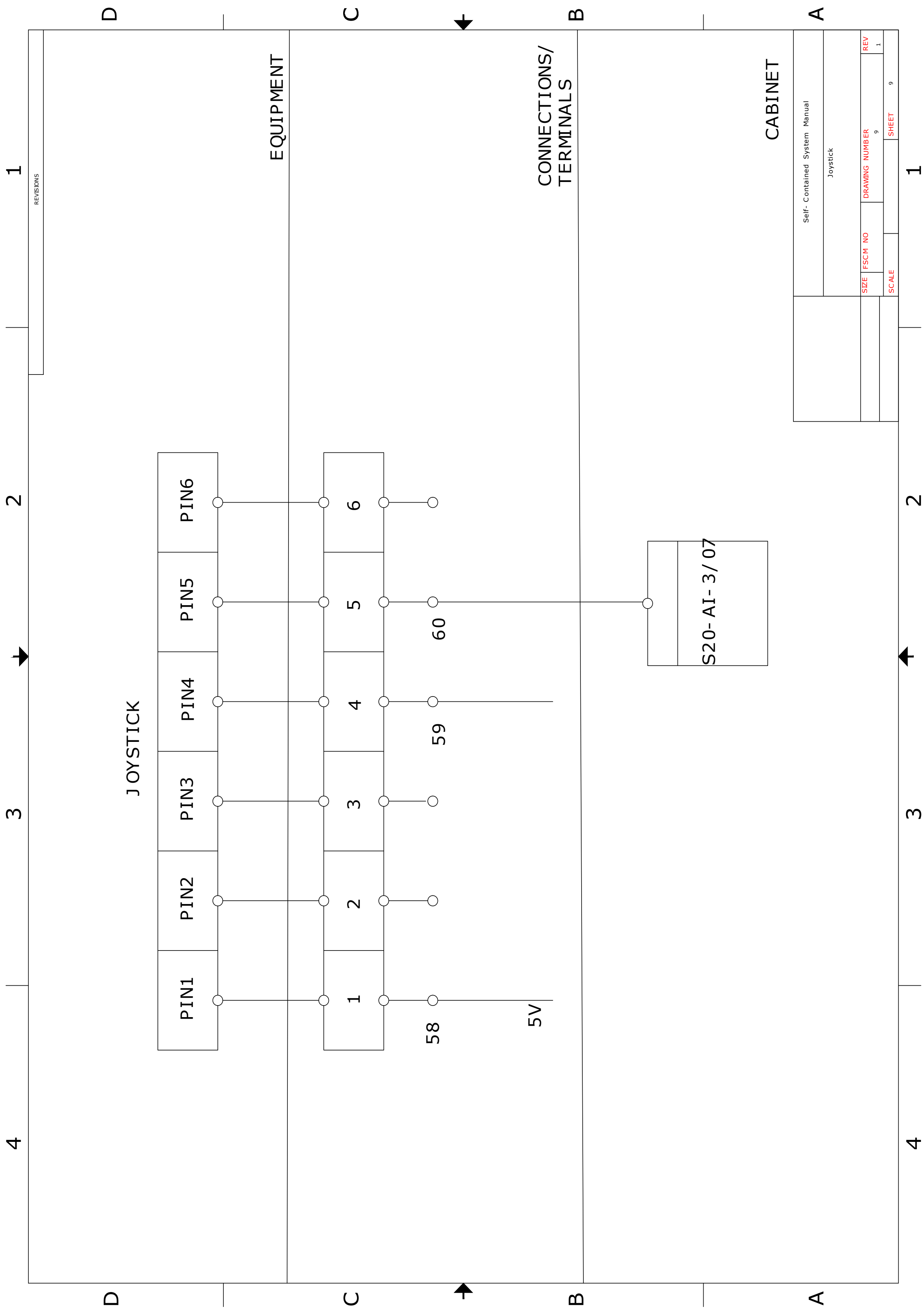

REVISIONS
3.054
TEMP SENSOR
4-20mA
-50-200oC
3.053
TEMP SENSOR
4-20mA
-50-200oC
3.052
TEMP SENSOR
4-20mA
-50-200oC
3.051
TEMP SENSOR
4-20mA
-50-200C
T
EQUIPMENT
CONNECTIONS/
TERMINALS
GND
24V
31
32
29
30
27
28
25
26
S20-AI-8/27
S20-02
S20-AI-8/26
S20-02
S20-AI-8/25
S20-02
S20-AI-8/24
S20-02
CABINET
Self-Contained System Manual
Temprature Sensor
SIZE
FSCM NO
DRAWING NUMBER
4
REV
1
SCALE
SHEET
4
A
B
C
D
1
2
3
4


REVISIONS
3.058
TEMP SENSOR
0-24mA
-50-200oC
T
3.057
TEMP SENSOR
0-24mA
-50-200oC
T
3.056
TEMP SENSOR
0-24mA
-50-200oC
T
3.055
TEMP SENSOR
0-24mA
-50-200oC
T
EQUIPMENT
GND
39
40
24V
37
38
35
36
33
34
CONNECTIONS/
TERMINALS
S20-AI-8/23
S20-03
S20-AI-8/22
S20-03
S20-AI-8/21
S20-03
S20-AI-8/20
S20-02
CABINET
Self-Contained System Manual
Temprature Sensor
SIZE
FSCM NO
DRAWING NUMBER
6
REV
1
SCALE
SHEET
6

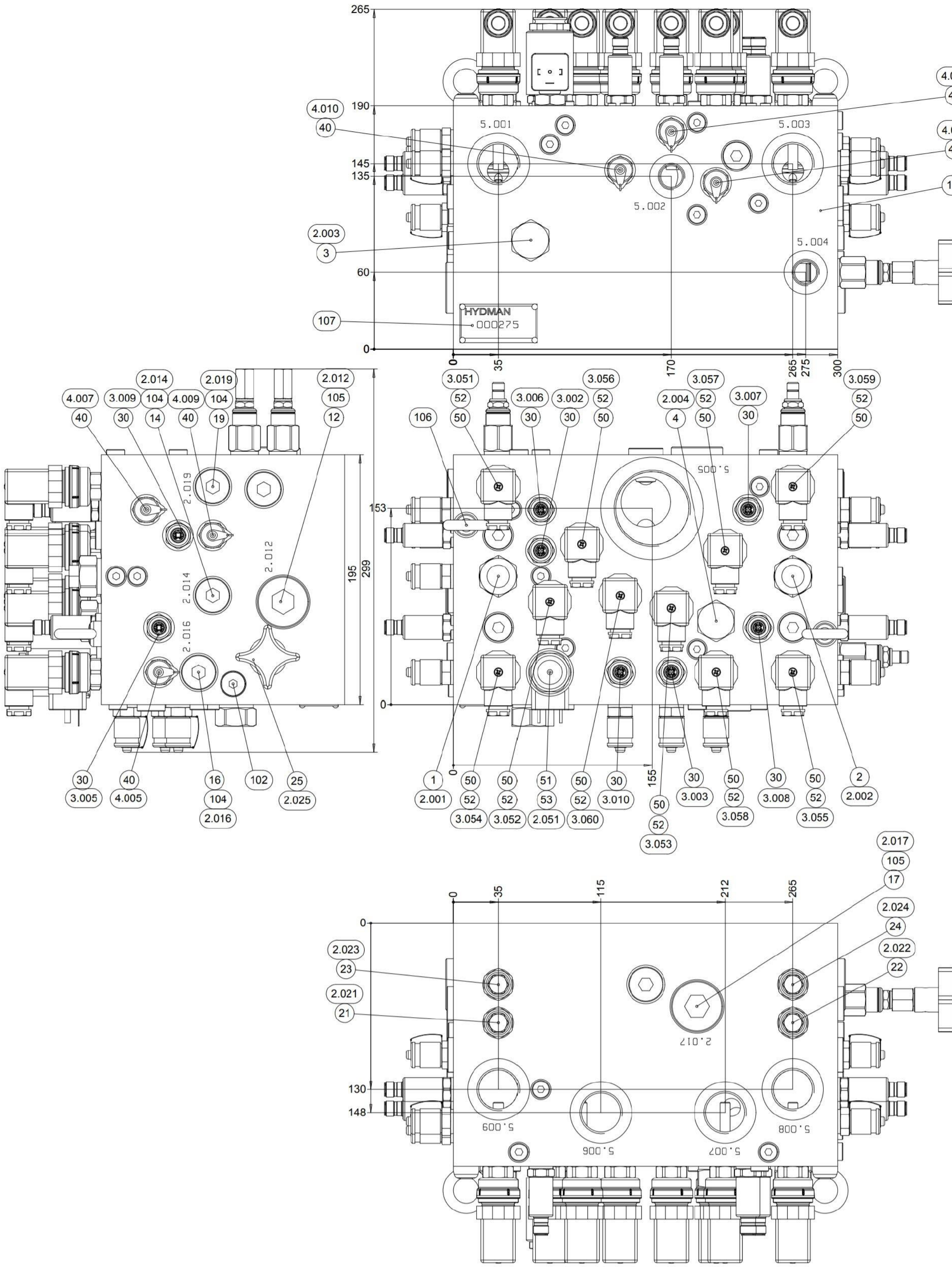


REVISIONS
TEMP SENSOR 3.062
0-24mA
-50-200₀C
TEMP SENSOR 3.061
0-24mA
-50-200₀C
TEMP SENSOR 3.060
0-24mA
-50-200₀C
TEMP SENSOR 3.059
0-24mA
-50-200C
T
EQUIPMENT
GND
24V
47
48
45
46
43
44
41
42
CONNECTIONS/ TERMINALS
S20-AI-8/27 S20-03
S20-AI-8/26 S20-03
S20-AI-8/25 S20-03
S20-AI-8/24 S20-03
CABINET
Self-Contained System Manual
Temprature Sensor
SIZE
FSCM NO
DRAWING NUMBER
REV
SCALE
SHEET
A
B
C
D
1
2
3
4


TURN BUTTOM
RED BUTTOM
GREEN BUTTOM
EQUIPMENT
CONNECTIONS/
TERMINALS
CABINET
24V
GND
X1
X2
3
4
65
64
66
63
03
02
S20- DI-
S20- DI- 01
S20- DI- 00
REVISIONS
Self- Contained System Manual
Push Buttons
SIZE
FSCM NO
DRAWING NUMBER
8
REV
1
SCALE
SHEET
8

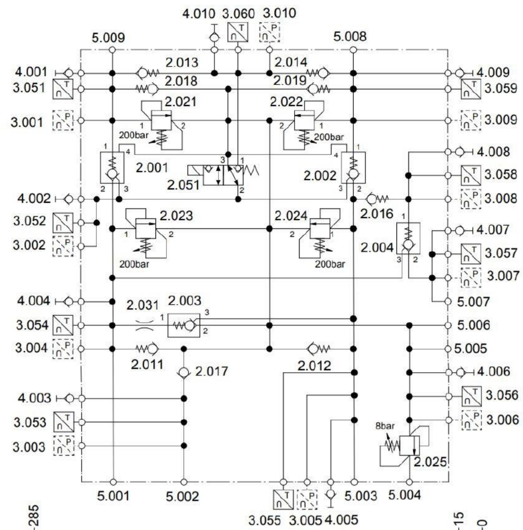


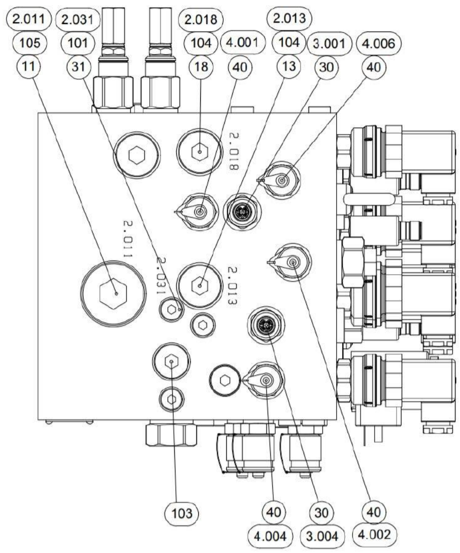
EQUIPMENT
CONNECTIONS/
TERMINALS
CABINET
JOYSTICK
PIN1
PIN2
PIN3
PIN4
PIN5
PIN6
1
2
3
4
5
6
58
59
60
5V
S20-AI-3/07
REVISIONS
Self-Contained System Manual
Joystick
SIZE
FSCM NO
DRAWING NUMBER
9
REV
1
SCALE
SHEET
9

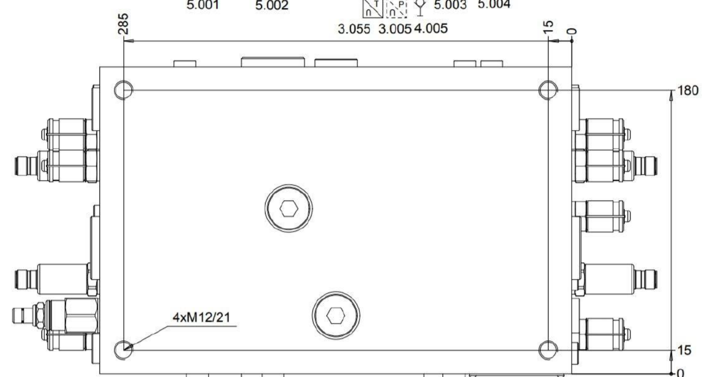

4.003
40
4.010
40
4.008
40
100
2.003
3
107
HYDMAN
000275
5.001
5.002
5.003
5.004
265
190
145
135
60
0
35
170
275
300
2.014
2.019
2.012
4.007
3.009
104
4.009
105
40
30
14
19
12
195
299
30
3.005
4.005
16
104
2.016
102
25
2.025
3.051
52
50
3.006
3.002
3.056
2.004
3.057
3.007
3.059
106
5.005
153
155
1
2.001
3.054
3.052
51
53
2.051
3.060
3.010
3.053
3.003
3.058
3.008
3.055
2
2.002
2.017
105
17
2.024
24
2.022
22
2.023
23
2.021
21
115
212
130
148
5.006
5.007
5.008
5.009

CONNECTIONS	
5.005	G 2
5.001, 5.003, 5.006	G 1
5.007, 5.008, 5.009	G1
5.002, 5.004	G1/2

MAX WORKING PRESSURE	210bar
STATIC PRESSURE TEST	315bar



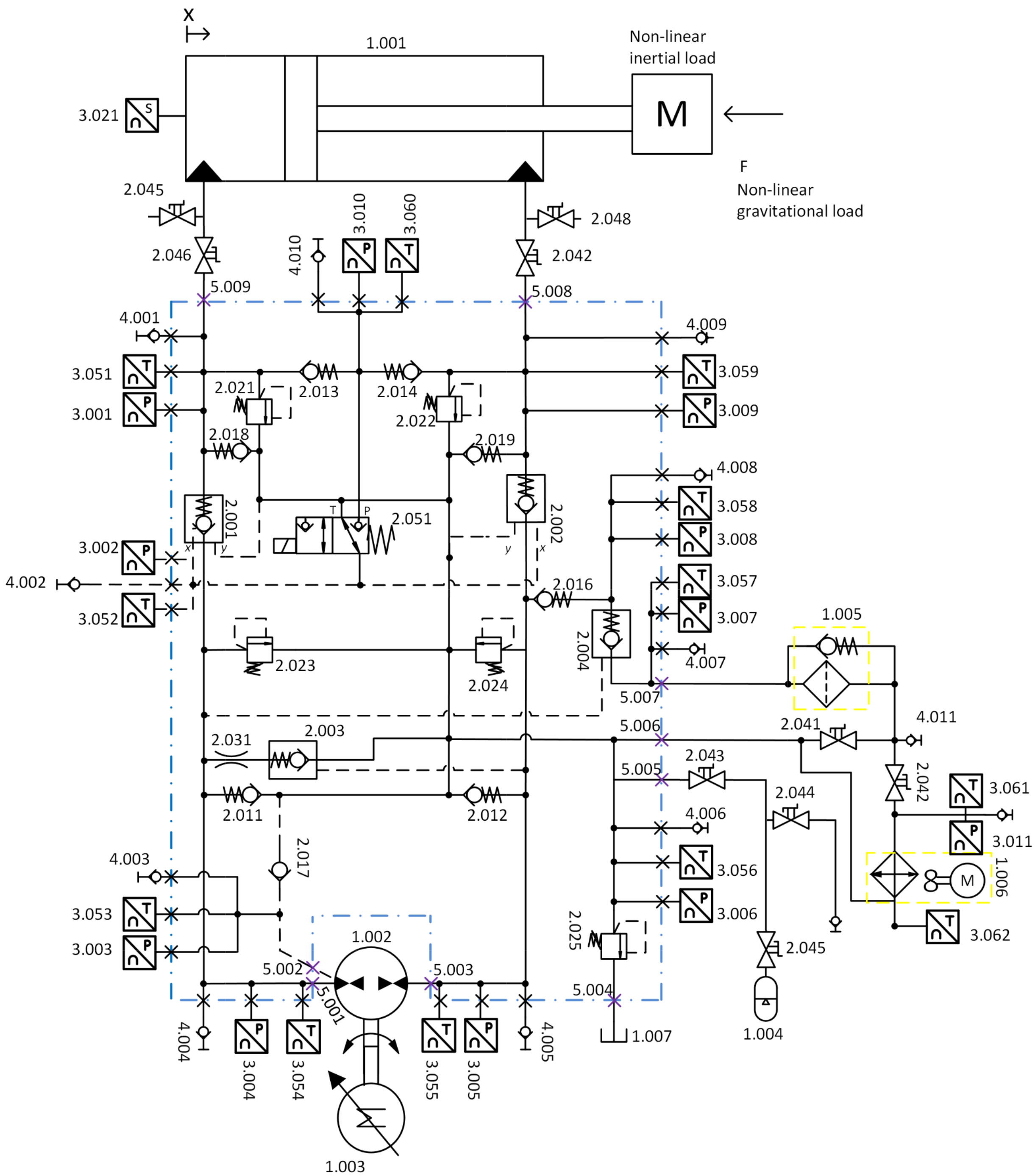


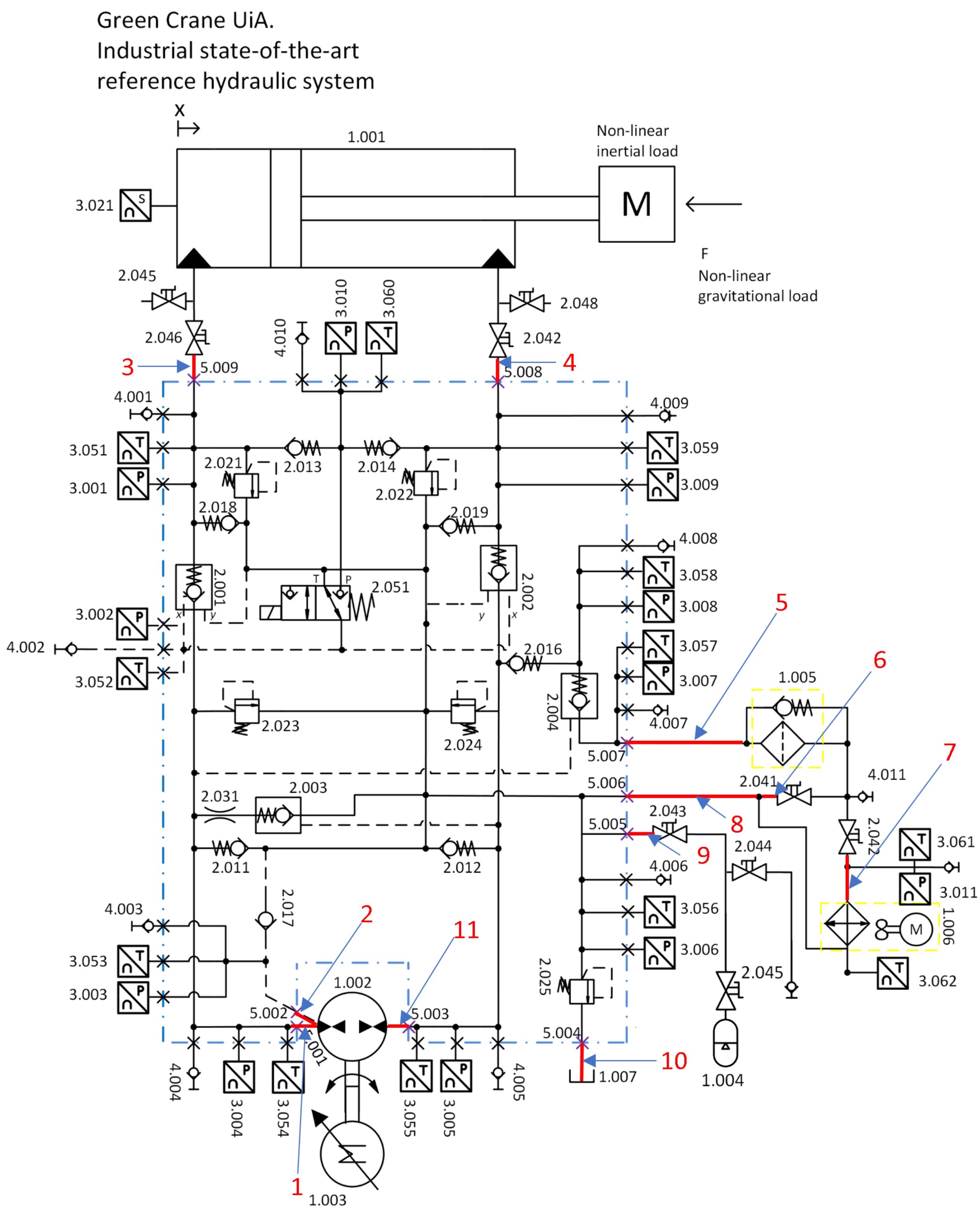


Part	Code	Description	Type	Mass	Qty
107	HM40016000	LABEL	HYDMAN 30x60		1
106	NO08000000	LIFTING EYE BOLT	M8 DIN580 ST	0.05	2
105	TUL1000000	PLUG	G 1" A		3
104	TUL0120000	PLUG	G 1/2" A	0.04	10
103	TUL0380000	PLUG	G 3/8" A	0.03	6
102	TUL0140000	PLUG	G 1/4" A	0.02	2
101	TUL0180000	PLUG	G 1/8" A	0.01	18
100	HM00027500	HYDRAULIC SYSTEM MANIFOLD	000275	25.81	1
53	KEAH000005	COIL	C19B-02400E1-25,75NA 27670600	0.20	1
52	US00400000	USIT SEAL	U/SC 13,74x20,57x2,0 G1/4		10
51	DIRAH34015	3/2- DIRECTIONAL POPPET VALVE	SD1E-A3/H2S8M9	0.22	1
50	STDA000008	TEMPERATURE TRANSDUCER 4-20mA	MBT 3560-0000-0050-10-110 084Z4030		10
40	MITST14000	MEASURING CONNECTOR	SMK20-G1/4-PC	0.08	10
31	KUR0080080	ORIFICE	M8-0.8		1
30		PRESSURE TRANCDUCER (NOT INCLUDED)	HM20-2X/...		10
25	RELSU10041	PRESSURE RELIEF VALVE	RDDA LSN + HAND KNOB	0.18	1
24	RELSU10019	PRESSURE RELIEF VALVE	RDDA LAN + CAPNUT 400158	0.17	1
23	RELSU10019	PRESSURE RELIEF VALVE	RDDA LAN + CAPNUT 400158	0.17	1
22	RELSU10019	PRESSURE RELIEF VALVE	RDDA LAN + CAPNUT 400158	0.17	1
21	RELSU10019	PRESSURE RELIEF VALVE	RDDA LAN + CAPNUT 400158	0.17	1
19	CHEHA02001	CHECK VALVE	RK2		1
18	CHEHA02001	CHECK VALVE	RK2		1
17	CHEHA04001	CHECK VALVE	RK4		1
16	CHEHA02000	CHECK VALVE	RB2		1
14	CHEHA02000	CHECK VALVE	RB2		1
13	CHEHA02000	CHECK VALVE	RB2		1
12	CHEHA04001	CHECK VALVE	RK4		1
11	CHEHA04001	CHECK VALVE	RK4		1
4	CHESU02001	PILOT TO OPEN CHECK VALVE	CKEB XCN	0.24	1
3	CHESU02001	PILOT TO OPEN CHECK VALVE	CKEB XCN	0.24	1
2	CHESU22001	PILOT TO OPEN CHECK VALVE	CVEV XFN	0.25	1
1	CHESU22001	PILOT TO OPEN CHECK VALVE	CVEV XFN	0.25	1

Assembly		Scale	General tolerances ISO2768-m					HYDRAULIC SYSTEM MANIFOLD
Mass	29.6 kg	1:2.5	Measure	>0.5-6	>6-30	>30-120	>120-400	>400-1000
			Tolerance	± 0.1	± 0.2	± 0.3	± 0.5	± 0.8
Design	2018-01-15 TN							
Check	2018-02-16 JH		HYDMAN					HM202036
Acc								Rev

Green Crane UiA.
Industrial state-of-the-art
reference hydraulic system




Green Crane UiA.
Industrial state-of-the-art
reference hydraulic system
x
1.001
Non-linear
inertial load
M
F
Non-linear
gravitational load
3.021
2.045
2.046
2.048
2.042
3.010
3.060
4.010
5.009
5.008
4.001
4.009
3.051
3.001
3.059
3.009
2.021
2.013
2.014
2.022
2.018
2.019
2.001
2.002
2.051
4.008
3.058
3.008
3.057
3.007
3.002
4.002
3.052
2.016
2.004
2.023
2.024
4.007
1.005
5.007
5.006
2.041
4.011
2.031
2.003
5.005
2.043
2.047
2.011
2.012
2.044
4.006
3.061
3.011
1.006
2.017
4.003
3.053
3.003
3.056
3.006
2.025
2.045
3.062
1.002
5.002
5.003
5.004
1.007
1.004
4.004
3.004
3.054
3.055
3.005
4.005
1.003
1
2
3
4
5
6
7
8
9
10
11

Item			
Number	From - To	Description	Value
1	Pump - Manifold	Pump Thread:	1” SAE
		Manifold thread	1” BSP
		Converter	1” BSP - 1” Metric
		Pipe Diameter	25mm
		Pipe Length	450 mm
		Pipe working Pressure	475 bar
2	Pump - Manifold	Pump Thread	9/16” ORFC
		Manifold thread	1/2” BSP
		Converter	9/16” ORFC - 1/2” Metric
		Hose Diameter	1/2”
		Hose Length	450 mm
		Hose working pressure	350 bar
3	Manifold - Cylinder	Manifold thread	1” BSP
		Cylinder	1/2” BSP
		Quick connections	1”
		Hose Diameter	1”
		Hose Length	800 mm
		Hose working pressure	350 bar
4	Cylinder - Manifold	Manifold Thread	1” BSP
		Cylinder Thread	1/2” BSP
		Quick connections	1”
		Hose Diameter	1”
		Hose Length	1000 mm
		Hose working pressure	350 bar
5	Manifold - Filter	Manifold Thread	1” BSP
		Filter Thread	3/4” BSP
		Hose Diameter	1”
		Hose Length	1000 mm
		Hose working pressure	175 bar
6	Filter - Manifold T	Filter Thread	3/4” BSP
		Manifold T Thread	3/4” BSP
		Hose Diameter	3/4”
		Hose Length	600 mm
		Hose working pressure	175 bar
		Hose working pressure	175 bar

Item			
Number	From - To	Info	Info
7	Filter - Cooler	Filter Thread	3/4” BSP
		Cooler Thread	3/4” BSP
		Hose Diameter	3/4”
		Hose Length	500 mm
		Hose working pressure	45 bar
8	Filter/Cooler - Manifold	Cooler Thread	3/4” BSP
		Manifold Thread	1” BSP
		Hose Diameter	1”
		Hose Length	1200 mm
		Hose working pressure	175 bar
9	Manifold - Accumulator	Manifold Thread	2” BSP
		Accumulator Thread	2” BSP
		Hose Diameter	1”
		Hose Length	1000 mm
		Hose working pressure	175bar
10	Manifold - Tank	Manifold Thread	1/2” BSP
		Tank Thread	None
		Pipe Diameter	1/2”
		Hose Length	500mm
11	Manifold - Pump	Manifold Thread	1” BSP
		Pump Thread	1”SAE
		Pipe Diameter	25mm
		Pipe Length	450mm
		Pipe working pressure	475 bar

A Components in the electrical cabinet

Components:	Manufacturer:	Art.Nr:	Quantity:
Cabinet (600x760x35)	Rittal	150-03-173	1
Wall mount cabinet	Rittal	150-18-064	1
Filteran	Stego	300-63-977	1
Inntackefilter	Stega	300-63-986	1
Thermostat	Pentair Schroff	110-90-502	1
400v Fuse	Schneider Electric	110-83-264	1
230v Circuit Breakers	Schneider Electric	110-83-497	1
24v Circuit Breakers	Schneider Electric	110-83-841	2
24v Power Supply	Mean Well	300-44-175	1
PLC IO relay	Phoenix Contact	137-21-570	5
Safety reley	Phoenix Contact	300-85-308	1
Emergency stop button	Siemens	300-21-233	1
Contatc guard rail	Pentair Schroff	110-15-525	1
Duckts	HellermannTyton	300-62-505	1
DIN rail	Phoenix Contact	148-08-916	5
Fuse 5 x 20 A	RND Components	300-43-471	10
Terminals (Fuse)	Phoenix Contact	148-29-258	5
Terminals	Phoenix Contact	148-29-231	100
End holders	Phoenix Contact	148-29-279	12
End lid	Phoenix Contact	148-29-235	12
Kortslutningsbøyle	Phoenix Contact	148-29-280	1
Terminals 400v	Phoenix Contact	148-29-276	3
Terminals 400v	Phoenix Contact	148-29-277	1
Earth rail	Pentair Schroff	110-15-451	1
Terminals	Pentair Schroff	110-89-822	1
Terminals	Pentair Schroff	110-89-823	1
White twisted wire (100m)	Kabeltronik	155-79-891	1
Brown twisted wire (100m)	Kabeltronik	155-79-893	1
Green/Yellow twisted wire (pr m)	Huber+Suhner	155-40-252	15
Powercable 5 x6.00 mm^2	Helukabel	300-09-596	15
CEE-plugg Red 32 A/400 VAC,	Bals	143-36-009	1
End sleeve 0.75 mm2	RND Connect	300-66-678	2
End sleeve 6 mm2	RND Connect	300-66-680	1
Mini DisplayPort	Bandridge	110-70-357	1
Mini DisplayPort-adapter	Bandridge	110-70-372	1
Mini DisplayPort – HDMI-	Bandridge	110-70-370	1

Cabinet components

Terminal tags	Phoenix Contact	148-28-157	10
Cable sleeve (6…12mm)	RND Components	300-96-614	1
Cable sleeve (11…16mm)	RND Components	300-96-615	1
Cable sleeve (10…15mm)	RND Components	300-96-618	10
Cable sleeve (3…6.5mm)	RND Components	300-96-616	15
CEE-Wall plug red 32 A/400 VAC	Bals	143-36-304	1
CEE-connector red 32 A/400	Bals	143-36-003	1
Swithch of switch 11.5 kW 32 A	Siemens	300-84-172	1
Green/yellow twisted wire (pr m)	Helukabel	300-09-565	3
Shrink cabl shoe 10 mm^2		148-39-601	1
Terminals	Phoenix Contact	148-29-231	50
Powercord 3 x1.5 mm^2		300-09-591	20
Niter 1.5m2		300-66-681	1
Joystick		164-13-415	1
2-led shilded cable		155-81-699	25
3-led shilded cable		155-81-700	25
Shorting bail		148-29-282	2
Green push button		301-02-877	1
Rocker switch		301-02-903	1
Red push button		301-02-879	1
Contolcable 12 x 0.75 mm^2		155-81-678	8
Terminals Yellow/Green		148-28-956	20
Shrink tube green		300-48-586	15
Plugg-in-bridge		137-20-886	1
Cabel marking ”0”	Partex	136-83-455	2
Cable marking ”1”	Partex	136-83-463	2
Cable marking ”2”	Partex	136-83-471	2
Cable marking ”3”	Partex	136-83-489	2
Cable marking ”4”	Partex	136-83-497	2
Cable marking ”5”	Partex	136-83-505	2
Cable marking ”6”	Partex	136-83-513	2
Cable marking ”7”	Partex	136-83-521	2
Cable marking ”8”	Partex	136-83-539	2
Cable marking ”9”	Partex	136-83-547	2

Cabinet components